Where linear Vlasov theory of magnetized cross-beam energy transfer can be trusted: a sum-rule-based validity map against particle-in-cell simulations

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Abstract

Linear Vlasov theory of magnetized cross-beam energy transfer (MagCBET) is fast enough for inertial-fusion design, but has been compared with kinetic simulations only qualitatively. We map where it is quantitatively predictive, against every archived gain curve of a recent two-dimensional particle-in-cell (PIC) study (carbon and hydrogen plasmas, 0–80 MG, plasma wave vector from parallel to nearly perpendicular to the field, two polarizations) and against new reduced-scale simulations along the laser-intensity axis. Because the simulations sample a picosecond transient of a weakly damped ion-acoustic wave, gains at a fixed detuning cannot be compared with a steady-state theory; the detuning-integrated first-order transfer is protected by a causality sum rule, and we use it, calibrated once at zero field, as a pre-registered agreement metric (threshold: 30% of the unmagnetized transfer) with Holm-controlled classification and a mechanism attribution that separates failures of the comparison from failures of the theory. On all 64 archived cells the theory predicts the magnetic modification of the transfer to within 0.16 of the unmagnetized transfer; no cell crosses the threshold, and the residual grows with magnetization and is largest near perpendicularity. Compared pointwise, the same data would violate the theory in 24 of 73 field-scan points—established failures of the quasi-steady-state prerequisite, not of the theory. Along the intensity axis the integrated transfer doubles between 0.5 and $10^{15}\,\mathrm{W/cm^2}$, as predicted; at $2 \times 10^{15}\,\mathrm{W/cm^2}$ a pre-registered "failure" attributed to the nonlinear ion response proves to be an artifact of noise-seeded scattering in the seed's spectral window, a diagnosis whose pre-stated predictions then held on two fresh cells. A failed resolution validation shows that near-perpendicular magnetized transfer is far more sensitive to numerical resolution than unmagnetized transfer. The absolute coupling is under-predicted by a factor 1.40–2.04 that we leave calibrated.

1 Introduction

Magnetic fields are now a routine ingredient of laser-driven inertial confinement fusion (ICF). Imposed fields raise the ion temperature and yield of capsule implosions on OMEGA and the National Ignition Facility [Chang et al., 2011, Moody et al., 2022, Sio et al., 2023], and magnetized designs promise ignition at reduced laser energy [Perkins et al., 2017, Walsh et al., 2022]. Even without coils, Biermann-battery fields of order 10^2 T fill the laser entrance holes of hohlraums [Farmer et al., 2017], exactly where the inner and outer beam cones cross. Cross-beam energy transfer (CBET)—the exchange of power between crossing beams through a driven ion-acoustic quasimode [Randall et al., 1981, Kruer et al., 1996]—is used

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on purpose to tune the symmetry of indirect-drive implosions [Michel et al., 2009, Moody et al., 2012] and is a dominant loss in direct drive [Igumenshchev et al., 2012, Froula et al., 2012, Marozas et al., 2018]. Magnetization changes it directly, yet magnetized NIF hohlraum implosions are presently modelled with unmagnetized CBET gains plus tuned multipliers and saturation clamps [Strozzi et al., 2024]; Ref. [Moody and Shi, 2026] reports that matching the hot-spot shape of the magnetized experiments required a clamp about ten times larger.

A linearized Vlasov theory of magnetized CBET (MagCBET) [Moody and Shi, 2026] predicts how: the transfer between initially parallel-polarized beams falls, orthogonally polarized beams start to couple, the two electromagnetic eigenmodes acquire different gains, and new resonances (ion Bernstein, lower hybrid) appear when the plasma wave vector is nearly perpendicular to the field. Two-dimensional particle-in-cell (PIC) simulations confirm these trends qualitatively but leave quantitative differences unexplained [Shi and Moody, 2026]. For a practitioner the operational question is sharper than "does the theory agree": *in which regimes can the fast linear theory be inserted into a CBET model, and where must one fall back to kinetic simulation?*

Answering it requires more than overlaying curves. A PIC run measures a gain a few picoseconds after the beams overlap, whereas the theory describes a steady state; for the weakly damped ion-acoustic wave of a carbon plasma the two differ by orders of magnitude at any single detuning (Sec. 2.3). Simulations also contain background losses, finite-domain effects and numerical heating, and their quoted uncertainties are spreads over a few samples. A useful validity map therefore needs (a) an observable that a steady-state theory can legitimately predict for a finite-duration simulation, (b) a decision rule and a parameter grid fixed before the comparison, (c) uncertainty propagation and error-rate control over many cells, and (d) a separation of failures of the *comparison* (the simulation does not test the theory) from failures of the *theory's physics*.

This paper builds such a map. Our contributions are:

1. a comparison observable—the detuning-integrated first-order transfer—that is protected by a causality sum rule and is therefore insensitive to the interaction time and to the damping rate, and the demonstration that pointwise gains are not comparable for carbon plasmas;
2. a pre-registered validity map of a reconstructed Vlasov MagCBET baseline against all archived gain curves of Ref. [Shi and Moody, 2026] across field strength (0–80 MG), field orientation ($\theta_{kB} \approx 0°$–$89°$), electron temperature (0.4–1.2 keV, two species) and polarization, and against new reduced-scale simulations along the laser-intensity axis (0.5–$2 \times 10^{15}\,\mathrm{W/cm^2}$), with Holm-controlled classification;
3. a mechanism attribution that inspects comparison-validity prerequisites (quasi-steady state, background drift, pump depletion, signal-to-noise, finite-gain nonlinearity) before the theory's assumptions (ponderomotive drive, linear ion response, single-quasimode response along unperturbed orbits);
4. diagnostics of when PIC benchmarks stop testing the theory—resolution sensitivity near perpendicularity, and noise-seeded scattering and pump depletion at high intensity, the latter diagnosis supported by pre-stated predictions on two fresh cells;
5. a compactness analysis identifying the dimensionless parameters that carry the boundary, and guidance for ICF practice.

2 Linearized Vlasov theory of magnetized cross-beam transfer

2.1 Configuration

We consider the configuration of the two-dimensional, three-velocity (2D-3V) simulations analysed below (Fig. 1). A uniform, collisionless, fully ionized plasma of electron density n_e and temperatures $T_e = 4T_i$ fills a slab $0 < x < L_x$, $|y| < L_y/2$ with $L_x = 500\lambda_0$, $L_y = 10\lambda_0$ and $\lambda_0 = 0.351\,\mu$m. A seed beam of vacuum wavelength $\lambda_1 = \lambda_0 + \Delta\lambda$ propagates along $+\hat{\mathbf{x}}$; a plane-wave pump of wavelength λ_0 crosses it at a nominal angle α (30° for carbon, 24° for hydrogen). The pump is launched from the $y = +L_y/2$ boundary with the phase of a vacuum wave, so that inside the plasma its wave vector is fixed by phase matching,

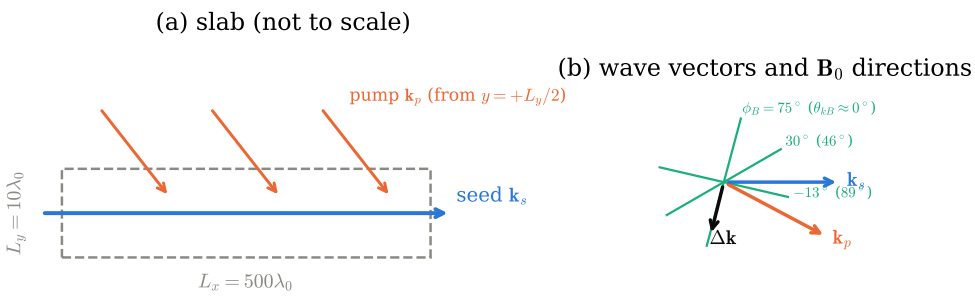


Figure 1: Geometry of the simulations. (a) Slab of $500\lambda_0 \times 10\lambda_0$ with the seed along $+x$ and a plane-wave pump launched from the upper boundary. (b) In-plasma wave vectors (computed by the solver; the pump is refracted to 27.96°) and the three in-plane field directions of the carbon cases with the resulting angle θ_{kB} between the plasma wave vector $\Delta\mathbf{k}$ and $\mathbf{B}_0$.

$k_{p,x} = (\omega_0/c)\cos\alpha$ and $k_{p,y} = -[n_p^2\omega_0^2/c^2 - k_{p,x}^2]^{1/2}$; for the carbon plasma this gives an in-plasma crossing angle of 27.96° (hydrogen: 21.71°). The beat between the beams drives a density perturbation at $(\Delta\mathbf{k}, \Delta\omega) = (\mathbf{k}_p - \mathbf{k}_s,\ \omega_0 - \omega_1)$; for $\Delta\lambda > 0$ energy flows to the seed. A uniform field $\mathbf{B}_0$ has polar angle θ_B and azimuth ϕ_B in the simulation frame. For the carbon cases ($\theta_B = 90°$) the three azimuths $\phi_B = 75°, 30°, -13°$ put the plasma wave vector at $\theta_{kB} \equiv \angle(\Delta\mathbf{k}, \mathbf{B}_0) = 1.1°$, 46.1° and 89.1° in the refraction-consistent geometry (nominally 0°, 45°, 88°); the hydrogen cases ($\theta_B = 45°, \phi_B = 15°$) have $\theta_{kB} = 72°$. Pump and seed are launched linearly polarized along $\hat{\mathbf{z}}$ ("z-z") or with the seed along $\hat{\mathbf{y}}$ ("z-y").

2.2 The reconstructed theory

We reconstruct the steady-state linearized Vlasov–Maxwell theory of magnetized CBET (MagCBET) [Moody and Shi, 2026] and apply it to this slab. All ingredients below are those of Ref. [Moody and Shi, 2026]; what is new is only the treatment of propagation (an exact coupled-mode solution in the slab instead of a sphere-averaged overlap factor) and the evaluation of the gain exactly as the simulation diagnostic measures it.

Laser eigenmodes. Each beam is a superposition of the two cold-plasma eigenmodes of $\mathsf{D}(n)\cdot\hat{\mathbf{e}} = [n^2(\hat{\mathbf{k}}\hat{\mathbf{k}} - \mathsf{I}) + \mathsf{K}(\omega)]\cdot\hat{\mathbf{e}} = 0$, with K the Stix dielectric of electrons and ions; X denotes the fast root (smaller n^2), O the slow root. Polarization vectors are taken transverse and orthonormal, as in the eigenmode analysis of the simulations (below 10 T, where the two roots coincide to numerical precision, a circular basis is used, as in the original analysis code). The seed field is $\mathbf{E}_s = \sum_m A_m(x)\hat{\mathbf{e}}_m e^{ik_m x}$ ($m \in \{\mathrm{X}, \mathrm{O}\}$) and the pump $\mathbf{E}_p = \sum_p B_p\hat{\mathbf{e}}_p e^{i\mathbf{k}_p\cdot\mathbf{r}}$, with B_p fixed by the launch polarization; the pump polarization is evaluated on the seed axis.

(i) Ponderomotive drive. The electrons feel the beat potential energy

$$\tilde{U} = \frac{e^2}{4m_e\bar{\omega}^2}\sum_{p,m} B_p A_m^* \left(\hat{\mathbf{e}}_p\cdot\bar{\mathsf{M}}\cdot\hat{\mathbf{e}}_m^* + \hat{\mathbf{e}}_m^*\cdot\bar{\mathsf{M}}\cdot\hat{\mathbf{e}}_p\right) e^{i(\mathbf{k}_p - k_m\hat{\mathbf{x}})\cdot\mathbf{r}}, \tag{1}$$

where $\bar{\mathsf{M}} = \mathsf{M}(\bar{\omega})$ is the cold magnetized quiver matrix ($\mathbf{v} = (iq/m\omega)\mathsf{M}\cdot\mathbf{E}$) evaluated at the single averaged frequency $\bar{\omega} = (\omega_0 + \omega_1)/2$. This is the electrostatic-like "potential" term of the magnetized ponderomotive force [Lee and Parks, 1983]; the magnetization term and the time-derivative terms are dropped. The approximation is controlled by $\Delta\omega \ll \Omega_{ce} \ll \omega_0$ [Moody and Shi, 2026]. Note that the symmetrized pair factor in Eq. (1) equals $2\hat{\mathbf{e}}_m^*\cdot\mathrm{Re}(\bar{\mathsf{M}})\cdot\hat{\mathbf{e}}_p$: the helicity (antisymmetric, $\mathcal{O}(\Omega_{ce}/\omega_0)$) part of $\bar{\mathsf{M}}$ cancels.

(ii)–(iii) Plasma response. The electron density response at the beat is $\delta\tilde{n}_e = -(\epsilon_0\Delta k^2/e^2)\,R(\Delta\mathbf{k},\Delta\omega)\,\tilde{U}$ with

$$R = \frac{\chi_e(1+\chi_i)}{1+\chi_e+\chi_i}, \qquad \chi_s = \frac{1}{k^2\lambda_{Ds}^2}\Big[1+\frac{\omega}{|k_\parallel|v_s}\sum_{n=-\infty}^{\infty}\Gamma_n(\lambda_s)\,Z\Big(\frac{\omega-n\Omega_s}{|k_\parallel|v_s}\Big)\Big], \tag{2}$$

the warm, fully magnetized Maxwellian (Harris) susceptibility obtained by integrating the linearized Vlasov equation along unperturbed helical orbits [Stix, 1992]; $v_s = (2T_s/m_s)^{1/2}$, $\Gamma_n = I_n(\lambda_s)e^{-\lambda_s}$, $\lambda_s = k_\perp^2 T_s/(m_s\Omega_s^2)$ and Z is the plasma dispersion function [Fried and Conte, 1961]. Equation (2) is the electrostatic limit of the full response of Ref. [Moody and Shi, 2026]; electromagnetic corrections are of order $(\omega/kc)^2 \lesssim 10^{-6}$ in the ion-acoustic window and the Alfvén and lower-hybrid resonances lie outside it ($\gtrsim 10\,\text{Å}$). The ions respond only to the self-consistent field (their direct ponderomotive drive, smaller by $Z^2 m_e/m_i \sim 10^{-3}$, and all nonlinear ion response are neglected): this is assumption (ii). Assumption (iii) is that the transfer is mediated by this linear, unperturbed-orbit response—for the parameters here a single ion-acoustic quasimode, which turns into a comb of ion-Bernstein resonances and an overdamped response as $\theta_{kB} \to 90°$ and $|\mathbf{B}|$ grows.

Propagation and the measured gain. The current $\mathbf{J} = -\frac{1}{2}e\,\delta\tilde{n}_e^*\mathbf{v}_p$ (pump quiver times density perturbation; ion current neglected) drives the seed eigenmodes,

$$\frac{dA_m}{dx} = i\,\Gamma_m u_m \sum_{m'} w_{m'} A_{m'} e^{i(k_{m'}-k_m)x}, \qquad \Gamma_m = \frac{\omega_1 e^2}{8k_m m_e^2 c^2 \bar{\omega}^2 \omega_0}, \tag{3}$$

with $u_m = \hat{\mathbf{e}}_m^* \cdot \mathsf{M}(\omega_0)\cdot\mathbf{E}_p$ and $w_{m'} = \sum_p \Delta k^2_{pm'} R^*_{pm'} c^*_{pm'} B^*_p$ ($c_{pm'}$ is the pair factor of Eq. (1)). The rank-one 2×2 system (3) is solved exactly; Faraday rotation of the seed enters through the phase mismatch $k_{m'}-k_m$. For a single mode it reduces to the textbook unmagnetized CBET growth rate $\kappa = \Gamma\Delta k^2|B|^2\,\mathrm{Im}\,R$, which our unit tests reproduce to machine precision (this normalization satisfies the Manley–Rowe relations and differs by a factor of two from our reading of the printed prefactor of Ref. [Moody and Shi, 2026]; the factor cancels in all calibrated comparisons). The predicted energy gain is computed exactly as the simulations measure it, $g = \ln[\int_0^{L_x}|\mathbf{E}_s|^2dx\,/\,\int_0^{L_x}|\mathbf{E}_s|^2dx|_{\text{no pump}}]$, together with the eigenmode gains $g_{\mathrm{X,O}}$ from $|A_{\mathrm{X,O}}|^2$.

2.3 What a steady-state theory can and cannot predict for a finite-duration simulation

For the carbon plasma ($ZT_e/T_i = 24$) the ion-acoustic wave is weakly damped: the root of $\epsilon(\Delta\mathbf{k},\omega)=0$ has $\gamma/\omega_r = 0.0095$, a damping time of 60 ps, while the simulations measure the gain only 2.44 ps after the pump is switched on. The steady-state gain curve is a resonance of half width ≈ 0.011 Å centred at 1.153 Å reaching $g \approx 37$; the simulated curve is 1.4 Å wide (full width at half maximum), peaks at $g - g_0 = 1.34$ near 0.9 Å and is still $g - g_0 = 0.38$ at $\Delta\lambda = 0$ (Fig. 2a). The simulation therefore samples a transient, not the steady state, and pointwise comparisons at fixed $\Delta\lambda$ test the line shape rather than the theory.

The frequency integral of the transfer is, however, protected. For a causal response $R(\omega) = \int_0^\infty g(\tau)e^{i\omega\tau}d\tau$ switched on at $t=0$, the response at time t is $R_t(\omega) = \int_0^t g(\tau)e^{i\omega\tau}d\tau = \pi^{-1}\int R(\omega')\,\sin[(\omega-\omega')t]/(\omega-\omega')\,d\omega'$, so that $\int \mathrm{Im}\,R_t\,d\omega = \int \mathrm{Im}\,R\,d\omega$ for every $t>0$, independently of the damping rate. Any observable *linear* in R inherits this sum rule. We therefore compare the integrated amplitude gain

$$T = \int_W G(\Delta\lambda)\,d\Delta\lambda, \tag{4}$$

over the sampled detuning window W, with $G = F^{-1}(e^{g-g_0})$ for the simulations ($F(G) = (e^G-1)/G$ converts energy gain to envelope gain for an exponential profile; g_0 is the background offset) and $G = 2g^{(1)}$ for the theory, where $g^{(1)}$ is the energy gain to first order in the pump intensity,

$$g^{(1)} = -\frac{2}{L_x\sum_m |A_m(0)|^2}\,\mathrm{Im}\sum_{m,m'} A_m^*(0)\,\Gamma_m u_m w_{m'}\,A_{m'}(0)\,\Phi(k_{m'}-k_m), \tag{5}$$

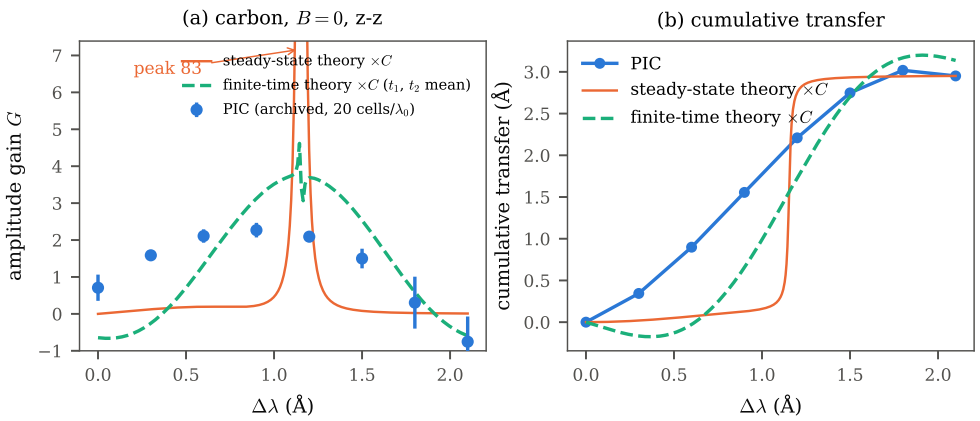


Figure 2: Why pointwise gains cannot be compared and integrated transfers can. (a) Carbon, $\mathbf{B} = 0$, z-z: archived simulation (points, offset removed, half-spread error bars), calibrated steady-state theory (narrow resonance, clipped) and calibrated finite-time theory at the measurement times (dashed). (b) Cumulative transfer $\int_0^{\Delta\lambda} G\, d\Delta\lambda'$; the three converge across the window although the line shapes differ by two orders of magnitude. The calibration constant C is fixed by the end point of the simulation and steady-state curves; the finite-time curve is a prediction.

with $\Phi(\delta) = (1 + i\delta L_x - e^{i\delta L_x})/\delta^2$. For a single gain mode $2g^{(1)} = F^{-1}(e^g)$ exactly; for mixed polarizations the first-order form is required because the steady-state exponentiation ($G \approx 41$ at the carbon resonance) would amplify tiny polarization projections and destroy linearity in R. Figure 2(b) illustrates the protection: the cumulative transfers of the simulation, the steady-state theory and the finite-time theory approach a common value across the window although their line shapes differ by two orders of magnitude.

2.4 Diagnostic variants used for attribution

Three variants of the theory isolate the assumptions; none is used as the baseline. (a) *Finite-time response*: $R \to R_t = R - \mathrm{Res}\, e^{i(\omega-\Omega)t}/(\omega - \Omega)$, keeping the weakly damped dispersion root Ω of $\epsilon = 0$ (when one exists) and evaluating at the measured interaction times; it relaxes the quasi-steady-state prerequisite. (b) *Helicity-retaining drive*: the pair factor of Eq. (1) is replaced by the sesquilinear form $\hat{\mathbf{e}}_m^* \cdot \bar{\mathsf{M}} \cdot \hat{\mathbf{e}}_p$ (and, separately, by the quiver form $(\mathsf{M}_0\hat{\mathbf{e}}_p) \cdot (\mathsf{M}_1\hat{\mathbf{e}}_m)^*$), which keeps the $\mathcal{O}(\Omega_{ce}/\omega_0)$ terms dropped by assumption (i). (c) *Ion ponderomotive drive*: the linear ion term omitted by assumption (ii), included to bound its size.

3 Simulation data

3.1 Archived simulations

The archived processed data [Shi, 2025] of the 2D-3V EPOCH [Arber et al., 2015] simulations of Ref. [Shi and Moody, 2026] form three configuration families (Table 1). *C-paper*: carbon ($Z = 6$, $n_e = 3.5 \times 10^{20}\,\mathrm{cm}^{-3}$, $T_e = 4T_i = 0.8\,\mathrm{keV}$), 20 cells per λ_0, 20 particles per cell, $I_p = I_s = 0.5 \times 10^{15}\,\mathrm{W/cm}^2$, the pump switched on when the seed has filled the box ($t_{\mathrm{pump}} = 0.60\,\mathrm{ps}$) and the gain measured between $t_0 = 0.61\,\mathrm{ps}$ and $t_{1,2} = 3.04, 3.64\,\mathrm{ps}$. Gain curves (8 detunings, 0–2.1 Å) exist for $|\mathbf{B}| \in \{0, 2, 5, 10, 50\}$ MG at each ϕ_B and both polarizations, plus z-y at 80 MG, $\phi_B = -13°$; $\Delta\lambda = 0.9$ Å scans cover 12 field values up to 64 MG. *H-T400* and *H-T1200*: hydrogen ($n_e = 3 \times 10^{20}\,\mathrm{cm}^{-3}$, $T_e = 4T_i = 0.4$ and $1.2\,\mathrm{keV}$, $\alpha = 24°$, $\theta_B = 45°$, $\phi_B = 15°$), whose archived input decks specify 10 cells per λ_0, 5 particles per cell, $I_p = I_s = 10^{15}\,\mathrm{W/cm}^2$, and an earlier timing protocol in which the pump is on from $t = 0$ and the gain is referenced to the first dump at 0.66 ps. Each archived gain carries

Table 1: Configuration families. Plasma: species, n_e ($10^{20}\,\mathrm{cm}^{-3}$), temperatures; resolution in cells per λ_0 / particles per cell; $I = I_p = I_s$ in $10^{15}\,\mathrm{W/cm^2}$; W is the detuning window of the integrated transfer.

Family	Plasma	α	resolution	I	W (Å)	cells
C-paper (archived)	C^{6+}, 3.5, $T_e = 4T_i = 0.8\,\mathrm{keV}$	30°	20/20	0.5	0–2.1	27 gain curves
H-T400 (archived)	H^+, 3.0, $T_e = 4T_i = 0.4\,\mathrm{keV}$	24°	10/5	1.0	0.1–2.2	26 gain curves
H-T1200 (archived)	H^+, 3.0, $T_e = 4T_i = 1.2\,\mathrm{keV}$	24°	10/5	1.0	0.1–2.2	17 gain curves
C-reduced (new)	C^{6+}, 3.5, $T_e = 4T_i = 0.8\,\mathrm{keV}$	30°	12/10	0.5, 1, 2	0–2.1	5 detunings/cell

Table 2: Numerical heating of a plasma-only carbon box (no lasers; electrons at t_1, ions at t_2) and cost of a full $500\lambda_0 \times 10\lambda_0$, 3.64-ps run, projected from short benchmarks without output (production runs with field output took 85 min at 12 and 125 min at 14 cells per λ_0).

cells/λ_0, ppc	$\delta T_e/T_e(t_1)$	$\delta T_i/T_i(t_2)$	s/step (ranks)	full run [h]
10, 10	10.3%	29%	0.040 (15)	0.51
12, 10	6.1%	25%	0.069 (15)	1.06
14, 10	3.9%	22%	0.079 (15)	1.42
20, 20	0.5%	7%	0.348 (16)	8.95

the documented spread between the two measurement times (t_1, t_2) and, for the eigenmode analysis, across the transverse integration fractions $f_y \in \{1, 0.75, 0.5\}$. The H-T400 files carry a zero spread (a single measurement time was used); those cells have no uncertainty band and are reported as illustrative only.

3.2 New reduced-scale simulations

The paper resolution costs ≈ 8.9 h per run on the 16 cores available to this study, so the new runs use a reduced configuration, held fixed for every new cell: the archived carbon input deck with 12 instead of 20 cells per λ_0 and 10 instead of 20 particles per cell; domain, timing, boundaries, species, output times and crossing geometry unchanged (85 min per run on 15 MPI ranks). The resolution was chosen by a rule fixed beforehand on plasma-only numerical-heating pilots (Table 2): the selected configuration heats electrons by 6.1% by t_1 and ions by 25% by t_2, against 0.5% and 7% at the paper resolution. Because a weakly damped ion wave is sensitive to such heating, the configuration had to pass a validation against the archived paper-resolution results before any new cell was evaluated (Sec. 5.6). To obtain reproducible yet distinct particle noise, we added a control-block key that offsets EPOCH's random seed (runs with the same seed are bitwise identical, `logs/determinism_check.txt`); repeats with three seeds at the two validation cells give a between-seed standard deviation of g at 0.9 Å of 0.062 at $\mathbf{B} = 0$ and 0.018 at 10 MG, no larger than the median half t_1–t_2 spread of the new runs (0.09); the pooled value $\sigma_{\mathrm{RNG}} = 0.046$ is added in quadrature to every new gain.

The new cells were planned along the laser-intensity axis, $I_p = I_s \in \{0.5, 1, 2\} \times 10^{15}$ W/cm^2, at $\mathbf{B} = 0$ and (at 0.5 and 2) at 10 MG with $\theta_{kB} = 89°$ (z-z), each with a gain curve at $\Delta\lambda \in \{0, 0.3, 0.9, 1.5, 2.1\}$ Å (a subset whose trapezoid reproduces the 8-point integrals of the archived curves to 0.05 Å rms) and a z-y, $\mathbf{B} = 0$ background run at every intensity. Because the reduced configuration failed its magnetized validation (Sec. 5.6), only the unmagnetized cells enter the map.

3.3 Gain extraction

The gain-extraction pipeline of Ref. [Shi and Moody, 2026] (Fourier band-pass of the seed, eigenmode reconstruction with Faraday phase alignment, energies at t_0, t_1, t_2, transverse fractions f_y) is available only in MATLAB; we ported it line by line to Python and verified it on synthetic fields with known exponential envelopes and elliptical eigenmodes, recovering total and eigenmode gains to within 0.004. The total gain used throughout is the Fourier-integral measure (robust for both polarizations); its uncertainty is half the larger of the documented spreads of the two measures. One robustness change was needed for the new

runs: the authors' spectral peak search, which walks from slightly above k_0 to the first local maximum, locked onto a spurious maximum in one dump of the z-y background run and produced an unphysical gain (+4.95). For every new run and dump we therefore keep the authors' integration window whenever it contains the seed peak (the maximum of the $k_y = 0$ power line within 0.9–1.1 k_0) and re-centre it on that peak otherwise; this quality-control fallback (fixed before any new-run residual was computed) was triggered in 4 of 108 dumps and is logged per run.

4 Quantifying agreement

All definitions in this section were fixed in a time-stamped plan before any theory–simulation residual was computed. The plan, its hashes and three amendments are part of the released material. The first two were made while evaluating the theory alone, before any residual existed: the first replaced the theory's nonlinear transfer by the first-order transfer of Eq. (5) (Sec. 2.3); the second added the seed-angle indicator A7 below. The third, made before any residual of the new simulations was computed, added a quality-control fallback to the gain-extraction pipeline (Sec. 3). Two later departures concern only the new simulations and are disclosed where they arise: the execution order was changed once, before any run at a new intensity existed, to complete the intensity axis first; and, after the reduced configuration failed its magnetized validation, its unmagnetized cells were evaluated without executing the fallback ladder (Sec. 5.6).

Background offset and calibration. Side- and back-scattering and the finite aperture make the measured seed energy fall even without coupling; z-y at $\mathbf{B} = 0$ has no CBET by symmetry, so its detuning-averaged gain defines the offset g_0 of each configuration family and intensity (C-paper: $g_0 = -0.234$; H-T400: -0.089; H-T1200: -0.188). Absolute coupling in a 2D slab with a diffracting Gaussian seed is not what a practitioner needs from MagCBET—CBET models are calibrated at $\mathbf{B} = 0$ against experiments in practice [Michel et al., 2009, Strozzi et al., 2017]—so each family f is calibrated once at its unmagnetized z-z anchor, $C_f = T_{\mathrm{PIC}}/T_{\mathrm{V}}$, and every other cell is a genuine prediction.

Residual and threshold. For a cell c of family f at intensity I,

$$\Delta_c = \frac{C_f T_{\mathrm{V}}(c) - T_{\mathrm{PIC}}(c)}{C_f T_{\mathrm{V}}(\mathbf{B} = 0, \text{z-z}; f, I)}, \tag{6}$$

the error of the calibrated prediction in units of the (linearly scaled) unmagnetized transfer. For magnetized cells Δ is the error of the predicted magnetic modification $\eta = T(\mathbf{B})/T(0)$; for unmagnetized cells at $I \neq I_{\mathrm{ref}}$ it is the departure from the linear-in-I scaling of the theory. We say the theory *holds* if $|\Delta| \leq \tau = 0.30$ and *breaks* otherwise; 0.30 corresponds to the level at which an ICF transfer multiplier would be retuned. $\tau = 0.2$ and 0.4 are reported only as sensitivity checks.

Uncertainty and inference. A parametric bootstrap (20 000 draws) samples every archived or new gain point from a normal distribution with standard deviation equal to half its documented spread, the offset of each family and intensity from its standard error (shared by all cells), and the anchors jointly, so that calibration uncertainty propagates; for the new runs the between-seed scatter of repeated runs is added in quadrature. Each label is *established* if the one-sided test in its direction ($H_0 : |\Delta| \geq \tau$ for "holds", $H_0 : |\Delta| \leq \tau$ for "breaks") rejects after Holm's step-down correction [Holm, 1979] over all tested cells (family-wise error rate 0.05), and *illustrative* otherwise. Choosing the direction from the point estimate does not inflate the error rate of a declared label, because the two rejection regions are disjoint. Cells whose spread derives from fewer than three samples (the 1.2-keV hydrogen archive, two measurement times) are flagged as thin and reported as illustrative even if their test rejects; they still count in the Holm family. Cells without a documented uncertainty (the 0.4-keV hydrogen archive) are not tested.

Mechanism indicators. Each "breaks" cell is attributed by a deterministic rule, fixed in advance, that inspects comparison-validity prerequisites before theory assumptions: A5

signal-to-noise (peak CBET signal $< 2\sigma$); A3 pump depletion (peak energy gain > 2); A6 finite-gain nonlinearity of the conversion $G = F^{-1}(e^g)$ for mixed polarizations (finite-time theory transfers, nonlinear vs first order, differ by $> \tau/2$); A2 background drift (z-z and z-y transfers equal within 2σ while both exceed the threshold, i.e. a polarization-blind loss change); A1 quasi-steady state (finite-time and steady-state normalized transfers differ by $> \tau/2$); A7 plane-wave idealization of the focused seed (theory averaged over the seed's angular spectrum, $\pm 4°$, changes η by $> \tau/2$; evaluated where $\theta_{kB} \geq 80°$); then B-i drive approximation ($\Omega_{ce}/\omega_0 \geq 0.1$ and the helicity-retaining drive moves the prediction by $> \tau/2$ toward the simulation); B-ii nonlinear ion response ($I \geq 10^{15}\,\mathrm{W/cm^2}$, transfer below the linear prediction by $> \tau$, ion bounce frequency in the driven wave exceeding the damping rate); B-iii magnetized response near perpendicularity ($\theta_{kB} \geq 70°$ and either $k_\perp \rho_e \leq 1$ or $\Omega_{ci}/\omega_r \geq 0.1$). A cell for which no indicator fires is "mechanism-unattributed". Four pre-registered tests address the mechanisms across cells (Holm over the family): M1, $|\Delta|$ larger at $\theta_{kB} \approx 89°$ than at $0°/46°$ for matched $(|\mathbf{B}|, \mathrm{pol})$ (exact Wilcoxon signed-rank); M2, sublinear transfer at $2 \times 10^{15}\,\mathrm{W/cm^2}$ (z test); M3, the helicity-retaining drive reduces $|\Delta|$ where $\Omega_{ce}/\omega_0 \geq 0.1$ (sign test); M5, single-detuning residuals exceed integrated residuals where $\gamma t_{\mathrm{int}} < 1$ (sign test).

Compactness. To ask which axes carry the boundary, we fit decision stumps (one variable) and depth-two trees (two variables) to the labels using the axes ($\log|\mathbf{B}|$, θ_{kB}, polarization, I, T_e) and six dimensionless parameters (seed Faraday phase $|k_\mathrm{O} - k_\mathrm{X}|L_x$, $k_\perp \rho_e$, Ω_{ci}/ω_r, Ω_{ce}/ω_0, γt_{int}, peak gain), score them by leave-one-out accuracy, and test the best single-variable rule against a max-statistic permutation null (10 000 permutations, which controls the error over all candidate variables [Westfall and Young, 1993]).

Multiplicity and outcome-dependent choices. Two families carry inferential claims, each controlled at family-wise error 0.05 by Holm's procedure: the per-cell labels (one one-sided test per tested cell, 43 tests) and the pre-registered mechanism tests (M1, M2, M3, M5; those that can be evaluated). The threshold $\tau = 0.30$ and the integrated metric were the only ones used for claims; $\tau = 0.2, 0.4$ are reported descriptively, and the single-detuning metric, considered and rejected before any residual was computed, appears only as the pre-registered secondary analysis entering M5. Analyses added after residuals were seen—the offset-uncertainty sensitivity, the compactness of the pointwise failures, the association of $|\Delta|$ with the pre-listed variables, and, for the new runs, the time histories, the Faraday check, the direct pump-depletion measurement, the pump-off and magnetized heating runs, the partial resolution-ladder step and the scattered-light correction—and a bootstrap-seed robustness check of the whole analysis are labelled exploratory; any p value quoted for them is corrected within the analysis (max-statistic permutation, or Holm over the pre-listed variables) and none supports a pre-registered claim. The one post-hoc mechanism we rely on, scattered light in the background measurement, was additionally tested against quantitative predictions stated, and hashed, before two fresh cells were simulated.

5 Results

5.1 The reconstruction reproduces the documented trends

Before any quantitative comparison we checked the reconstructed theory against the five qualitative trends reported for the archived simulations [Shi and Moody, 2026] (`logs/reproduction.json`). It reproduces 4 of 5: (1) the z-z transfer decreases monotonically with $|\mathbf{B}|$ at every orientation; (2) z-y transfer switches on with $|\mathbf{B}|$ at $\theta_{kB} \approx 0°$ and 46°; (4) at $\theta_{kB} = 89°$ the steady-state resonance broadens (full width 0.018 Å at $\mathbf{B} = 0$, 0.26 Å at 10 MG) and its maximum moves to 0.41 Å at 50 MG; (5) X- and O-mode transfers split (relative difference 0.026 at 2 MG, 0.12 at 10 MG, $\theta_{kB} \approx 0°$). It fails trend (3): near perpendicularity the theory predicts that z-y transfer reaches $\eta = 0.24$ of the unmagnetized z-z transfer at 10 MG, whereas the simulated z-y transfer stays ≤ 0.11—the seed propagates almost along $\mathbf{B}$ at $\phi_B = -13°$, so Faraday rotation mixes the polarizations in the theory more efficiently than in the simulations. This is the one place where a qualitative statement

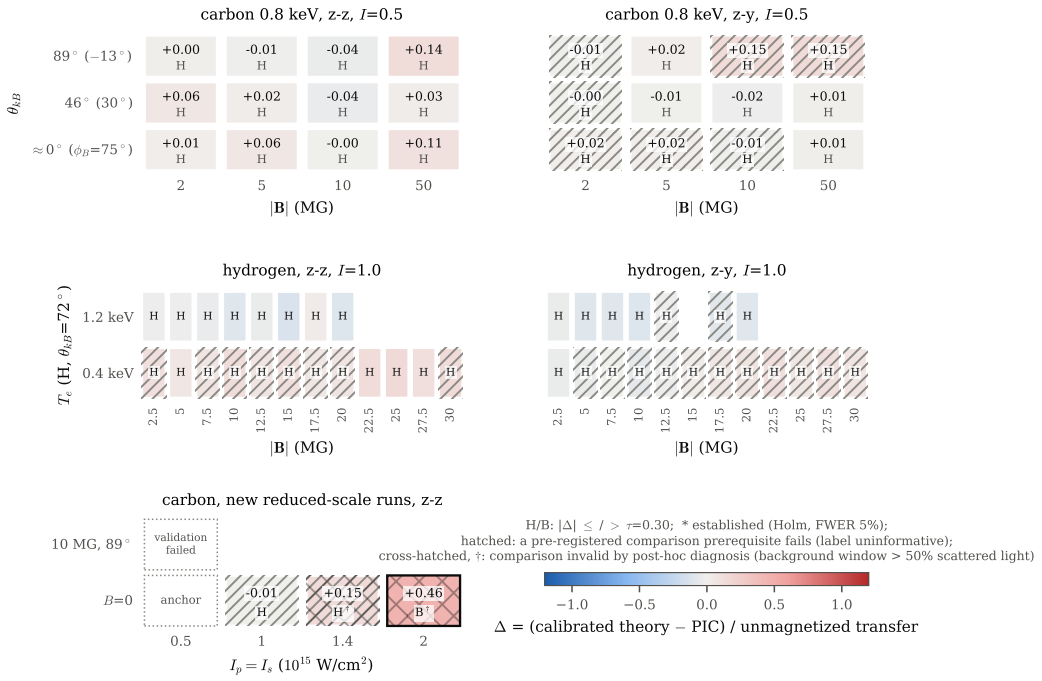


Figure 3: Validity map. Colour: agreement metric Δ (Eq. 6); letters: point label (H holds, B breaks at $\tau = 0.30$; * established under Holm control). Hatched: a pre-registered comparison prerequisite fails, so the label is uninformative; cross-hatched (†): the comparison is invalid according to the post-hoc diagnosis of Sec. 5.7. Top: archived carbon cells versus field strength and orientation θ_{kB}; middle: archived hydrogen cells versus field strength and temperature; bottom: new reduced-scale carbon cells versus laser intensity (dotted boxes: the calibration anchor and the magnetized validation cell, at which the reduced configuration failed).

of Ref. [Shi and Moody, 2026] and the theory disagree, and it reappears below as the largest residual of the carbon map.

5.2 Absolute coupling at $\mathbf{B} = 0$

The calibration constants are $C_f = 2.04$ (carbon), 1.40 (hydrogen, 0.4 keV) and 1.51 (hydrogen, 1.2 keV): the simulations transfer 1.4–2 times more than the first-order steady-state theory predicts for the same window. Several identified contributions are each too small to account for this alone: the finite-time response integrates to 1.06 times the steady-state value over the window (the transient sinc spills its first negative lobe partly outside); the factor-two ambiguity in the published ponderomotive normalization (Sec. 2) would make the discrepancy larger, not smaller; and the archived z-z curves are positive already at $\Delta\lambda = 0$ ($g - g_0 = 0.38$ for the carbon anchor), where at the measurement times neither the steady-state nor the finite-time linear response transfers energy; the new runs show that this gain accumulates along x like cross-beam transfer (Sec. 5.6), so it is a coupling of the beams that the linear response does not describe, and it grows with the coupling (it vanishes when CBET is suppressed at 50 MG, $\theta_{kB} = 89°$). We therefore treat the absolute level as unexplained and calibrated; all validity statements below concern the *magnetic modification* of a transfer whose unmagnetized value has been matched, which is how a MagCBET correction would enter an existing CBET model.

5.3 The validity map on the archived simulations

Table 5 lists every cell and Fig. 3 shows the map. Across all 64 archived test cells—carbon at 2–80 MG and $\theta_{kB} \approx 0°$, 46°, 89°, hydrogen at 2.5–30 MG and $\theta_{kB} = 72°$ at two temperatures,

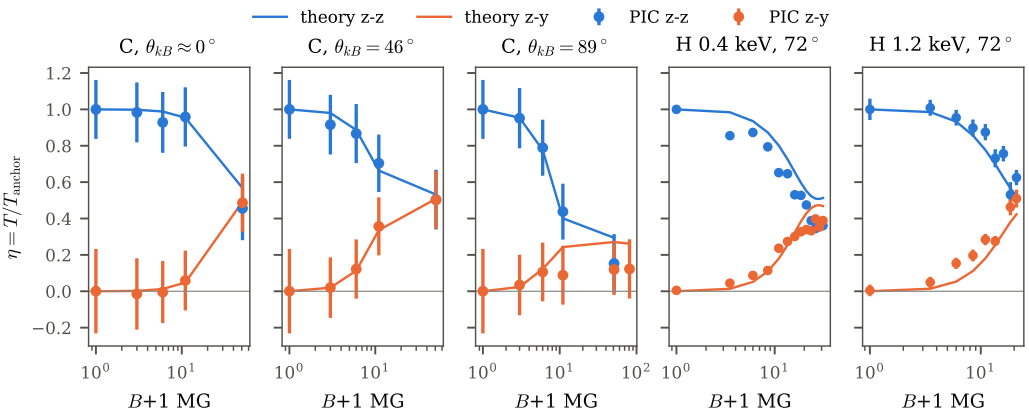


Figure 4: Magnetic modification of the integrated transfer, $\eta = T(\mathbf{B})/T(0)$, for z-z (blue) and z-y (orange): archived simulations (points, bootstrap s.d.) and first-order steady-state theory (lines, no free parameter once calibrated at $\mathbf{B} = 0$).

both polarizations—the calibrated theory stays inside the pre-registered band: $|\Delta| \leq 0.16$, with no cell above $\tau = 0.30$ (nor above 0.20). Figure 4 shows why: the predicted magnetic modification $\eta = T(\mathbf{B})/T(0)$ tracks the simulated one through suppression of z-z coupling (carbon, 89°, 10 MG: $\eta_{\mathrm{PIC}} = 0.44$, $\eta_{\mathrm{V}} = 0.40$; 46°, 10 MG: 0.70 vs 0.66), through activation of z-y coupling (46°, 50 MG: 0.50 vs 0.51; 0°, 50 MG: 0.49 vs 0.49) and in both hydrogen families. Carbon residuals have median $|\Delta| = 0.02$; hydrogen residuals are slightly larger and of opposite sign at the two temperatures (mean $\Delta = +0.07$ at 0.4 keV, where the theory over-predicts the field-induced reduction of z-z coupling, and -0.06 at 1.2 keV, where it under-predicts z-y activation).

None of these "holds" labels is *established* at family-wise error 0.05. For carbon the bootstrap spread of Δ (median 0.15) is dominated by the uncertainty of the shared background offset, whose documented spread is inflated by an artifact of the eigenmode analysis at z-y, $\mathbf{B} = 0$ (Appendix B); only 10 of 25 carbon cells have unadjusted 95% intervals inside $\pm\tau$, and the Holm-corrected equivalence tests cannot reject. The 1.2-keV hydrogen cells have tight intervals (15 of 15 inside $\pm\tau$) but their spreads come from only two measurement times and are therefore flagged as thin, and the 0.4-keV archive carries no spread at all. The map is thus a map of *consistent* cells: the data never contradict the theory at the 0.3 level, and they confirm it at that level only descriptively.

Not every consistent cell is an informative test. The pre-registered comparison prerequisites (Sec. 4) fail for 8 of the 25 carbon cells—all of them z-y cells at low field or near-zero coupling, where both theory and simulation transfer almost nothing (A5, signal-to-noise)—and for 19 of 24 and 2 of 15 hydrogen cells. In the 0.4-keV hydrogen archive the measured gains are large enough ($I = 10^{15}$ W/cm^2, peak energy gain $g - g_0$ up to 2.2) that, according to the finite-time theory, converting the measured energy gain to an envelope gain distorts the transfer of mixed-polarization fields by up to 0.43 of the unmagnetized transfer (A6), and at 30 MG the old timing protocol itself changes η by 0.21 (A1). These labels are hatched in Fig. 3: they agree with the theory, but the comparison does not test it. The 35 archived cells whose comparison is valid all hold, with $|\Delta| \leq 0.16$.

5.4 Where the residuals concentrate

Although no cell crosses the threshold, the residual magnitude is not random. Across all tested cells $|\Delta|$ correlates with every measure of field strength—Ω_{ci}/ω_r ($\rho = 0.45$, exploratory, Holm-adjusted $p = 0.0015$ over the 11 pre-listed variables), the seed Faraday phase ($\rho = 0.42$), $\log|\mathbf{B}|$ ($\rho = 0.40$), Ω_{ce}/ω_0 ($\rho = 0.40$)—and with θ_{kB} ($\rho = 0.38$), but not with polarization ($\rho = -0.21$) or temperature ($\rho = -0.16$). These field proxies are collinear in the archived grid, so the residual structure is one-dimensional: residuals grow with magnetization. In

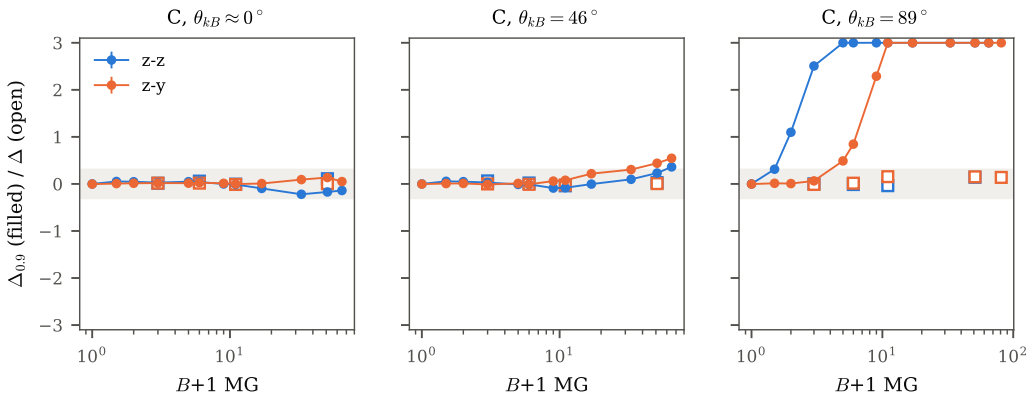


Figure 5: Pointwise versus integrated comparison for the carbon field scans. Filled: single-detuning residual $\Delta_{0.9}$ at $\Delta\lambda = 0.9$ Å (calibrated at the anchor's 0.9-Å point; clipped at ± 3); open squares: integrated residual Δ of the same cells; grey band: $\pm\tau$.

carbon the largest residuals are the near-perpendicular cells at ≥ 10 MG, where the theory over-predicts both z-z (50 MG: $\eta_{\mathrm{PIC}} = 0.15$, $\eta_{\mathrm{V}} = 0.29$) and z-y (10 MG: 0.09 vs 0.24) transfer by up to 0.15. The pre-registered test that residuals are larger at $\theta_{kB} \approx 89^\circ$ than at $0^\circ/46^\circ$ for matched $(|\mathbf{B}|, \mathrm{pol})$ (M1) gives $p = 0.125$ (Holm 0.25): suggestive, not established.

5.5 Pointwise comparisons fail for a reason that is not the theory

Had we compared gains at the detuning of the archived field scans (0.9 Å), the conclusion would have been the opposite (Fig. 5). Calibrating at the 0.9-Å point of the anchor requires a factor $C' = 20.6$ instead of $C = 2.04$, because the steady-state resonance puts almost no gain there; with that calibration $|\Delta_{0.9}| > \tau$ in 24 of the 73 archived field-scan points: 20 of 25 at 89°, 4 of 24 at 46° and 0 of 24 at 0°, with $|\Delta_{0.9}|$ up to 5.1. The failures concentrate where magnetization broadens the steady-state response (near-perpendicular geometry) and so redistributes theoretical gain toward 0.9 Å, while the simulated response—a finite-time transient of width $\sim 1/t_{\mathrm{int}}$—hardly changes shape. The pre-registered test M5 confirms that, in the weakly damped regime ($\gamma t_{\mathrm{int}} < 1$, all carbon cells), the single-detuning residual exceeds the integrated residual in 20 of 25 cells ($p = 0.00204$, Holm-adjusted 0.00815): these failures are failures of the quasi-steady-state prerequisite of the comparison (A1), not of the theory.

5.6 New simulations: validation of the reduced configuration

Before any new cell was evaluated, the reduced configuration had to reproduce two archived paper-resolution gain curves (Fig. 6). At $\mathbf{B} = 0$ it does: it transfers $T = 2.50$ against 2.95, a ratio of 0.85 ± 0.22 inside the pre-registered band $[0.75, 1.33]$ (V1), and the two curves differ pointwise by -0.19 to $+0.04$ in $g - g_0$. At 10 MG and $\theta_{kB} = 89^\circ$ it does not: the reduced runs keep a fraction $\eta = 0.77$ of the unmagnetized transfer, against 0.44 at the paper resolution (theory: 0.40), so V2 fails ($\Delta\eta = +0.33 \pm 0.18$ against a tolerance of 0.15), and so does the qualitative check V3 ($\eta < 0.6$). The reduced magnetized curve lies above the paper-resolution curve by up to $+0.28$ in $g - g_0$ at every detuning up to 1.5 Å. In the same configuration, then, the unmagnetized transfer is insensitive to resolution while the near-perpendicular magnetized transfer is not: going from 20 to 12 cells per λ_0 and from 20 to 10 particles per cell changes η at 89° by more than the largest theory–simulation residual of the archived map (Sec. 5.4).

Consequence for the campaign (disclosed deviation). The pre-registered response to a failed validation was to repeat the validation runs at 14 and then 16 cells per λ_0 and to use the first configuration that passes for every new cell. One rung of this ladder costs eleven runs—≈ 23 h at 14 and ≈ 37 h at 16 cells per λ_0—before any new cell, and the sweep

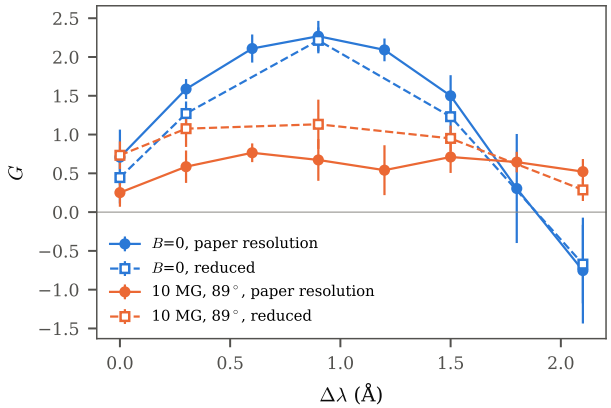


Figure 6: Validation of the reduced configuration: archived paper-resolution gain curves (filled) and reduced-scale curves (open) at $\mathbf{B} = 0$ and at 10 MG, $\theta_{kB} = 89°$ (envelope gain, offset removed, half-spread error bars).

at 14 cells per λ_0 a further ≈ 44 h, more than the remaining budget. We therefore deviated from the plan, after V2 had failed and before any run at a new intensity existed: because V1 validates exactly the unmagnetized transfer of the configuration, the unmagnetized intensity cells were run with the reduced configuration and are reported as validated by V1 only, while every magnetized new cell is treated as not validated and excluded from classification and from all claims. The planned magnetized gain curve at 2×10^{15} W/cm^2 and the magnetized refinement cells of the plan were therefore not run.

What the failure is not. It is not particle noise: with three random seeds the reduced runs give $g = 0.41$–0.45 at 10 MG and 0.9 Å (and 1.02–1.13 at $\mathbf{B} = 0$), so that $g - g_0 = 0.60$ exceeds the paper-resolution value 0.36 by ≈ 5 pooled between-seed standard deviations ($\sigma_{\mathrm{RNG}} = 0.046$). Bulk numerical heating does not explain it either: plasma-only pilots at the reduced resolution heat the electrons by 6.7% without field and 6.3% at 10 MG by t_1 (ions: 26% and 27% by t_2; `logs/pilot_heating_B.txt`), so the two validation cells are heated alike. Nor is it a field-dependent background loss of the seed: in an exploratory run with the pump switched off, the 10 MG seed loses $g = -0.16$, the same as in the z-y background run at $\mathbf{B} = 0$ (-0.17; dump by dump within 0.02), which also confirms that the unmagnetized offset applies to magnetized cells. The excess is therefore coupling: at 10 MG and 89° the reduced simulations transfer more energy than the paper-resolution ones. Its mechanism is not identified. The reduced grid resolves the 10-MG electron Larmor radius (67 nm) with 2.3 instead of 3.8 cells and the Debye length with 0.38 instead of 0.64 cells, and carries 5.6 times fewer particles per λ_0^2; grid- or noise-induced cross-field motion of the electrons would weaken the electron magnetization that suppresses the coupling near perpendicularity, in the observed direction. This post-hoc hypothesis was probed with one run of the pre-registered ladder at 14 cells per λ_0 (two restart segments; exploratory): at 0.9 and 0 Å it gives $g = 0.36$ and 0.14, against 0.43 and 0.22 at 12 and 0.12 and -0.11 at 20 cells per λ_0. Both detunings close the same fraction of the gap, 23% and 23%: the excess shrinks steadily but slowly with resolution, the reduced configuration is far from converged near perpendicularity, and, since the paper resolution also doubles the particle number per cell, nothing shows that it is converged there either. The seed polarization in the reduced magnetized runs follows the cold-plasma eigenmodes (Sec. 6.1), so the laser propagation is resolved.

Time histories. Because the new runs keep all six dumps, they show how the simulated gain develops (Fig. 7; exploratory). At $\mathbf{B} = 0$ and the resonant detuning the gain grows steadily, reaching $G = 2.37$ at the last dump against 2.82 for the finite-time theory and 0.19 for the steady state at the same detuning; on the red side (2.1 Å) simulation and finite-time theory first gain and then lose energy ($G = -0.89$ versus -0.73 at the last dump). The unmagnetized simulated curve is the transient of the linear response, as the

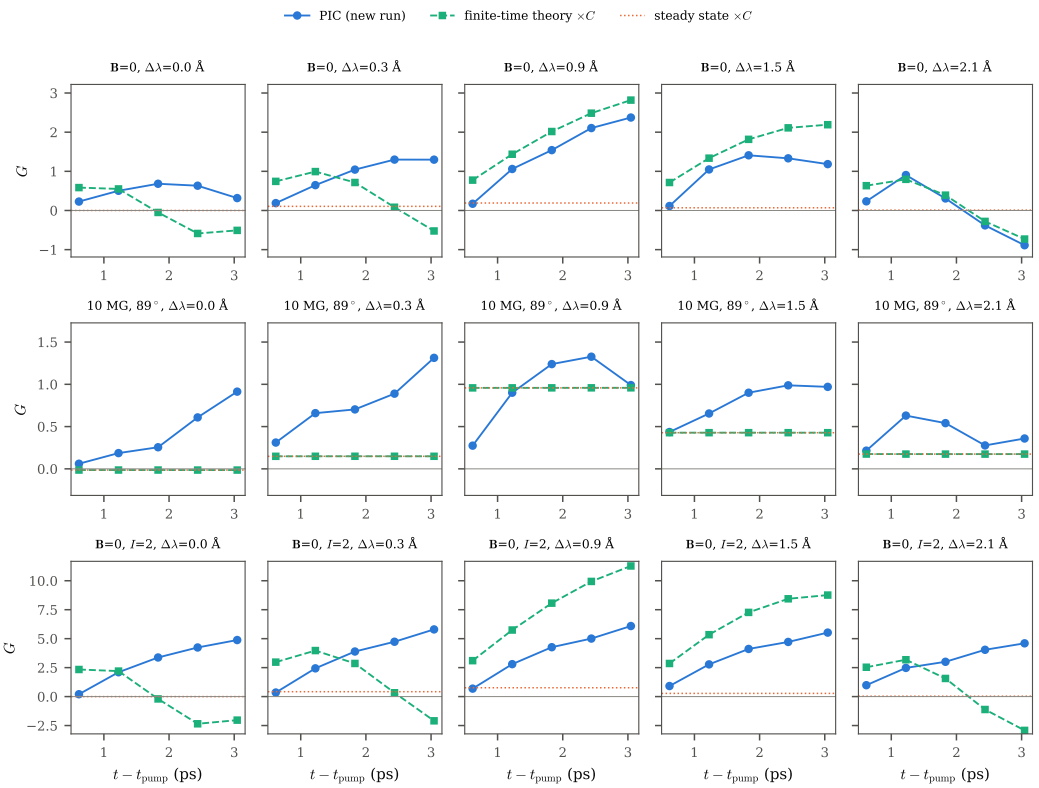


Figure 7: Time history of the envelope gain in the new reduced-scale runs (all six dumps; t_{pump} is the pump turn-on), compared with the finite-time linear response of the theory (calibrated with the reduced anchor constant) and with the steady state. Top: $\mathbf{B} = 0$; middle: 10 MG, $\theta_{kB} = 89°$ (configuration that failed validation V2 at this field); bottom: $\mathbf{B} = 0$ at 2×10^{15} W/cm^2. Offsets from the seed's own polarization in the background run. Exploratory analysis.

sum-rule argument assumes. The exception is again $\Delta\lambda = 0$, where the theory swings to loss ($G = -0.59$) while the simulated gain stays positive (up to 0.68); the seed's amplification accumulates along x there, so it is a coupling of the beams rather than a background artifact, but its origin remains open. At 10 MG the theory has no weakly damped quasimode and predicts a time-independent response, whereas the reduced simulations show gain that keeps growing at small detunings. The pump is not depleted in any of these runs: at the seed exit its intensity falls by at most 1.3% of the reference (`logs/pump_depletion.json`), because the focused seed of the archived deck carries only a few per cent of the pump power.

5.7 New simulations: the laser-intensity axis

At $\mathbf{B} = 0$ the first-order theory predicts an integrated transfer proportional to the pump intensity (the seed intensity drops out of Eq. 5). At 10^{15} W/cm^2 the pre-registered analysis finds $T_{\mathrm{PIC}} = 5.04 \pm 0.60$ against $C_f T_{\mathrm{V}} = 5.0$, i.e. $\Delta = -0.01 \pm 0.31$: the integrated transfer doubles with the intensity, as predicted, although the gain curve has broadened, most visibly on the blue side (Fig. 8a). By the pre-registered prerequisite A3 the label is nevertheless uninformative, because the peak energy gain exceeds 2 ($g - g_0 = 2.2$). That proxy presumes a seed as strong as the pump; measured directly (exploratory), the pump intensity at the seed exit falls by at most 2.0% (1.3% at the anchor intensity), because the focused seed of the archived deck carries only a few per cent of the pump power. Read with that measurement, the 10^{15} W/cm^2 cell supports the linear-in-intensity scaling of the integrated transfer—assumption (ii) at the level of the integrated observable—to within $|\Delta| \leq 0.06$ for every treatment of the background discussed below.

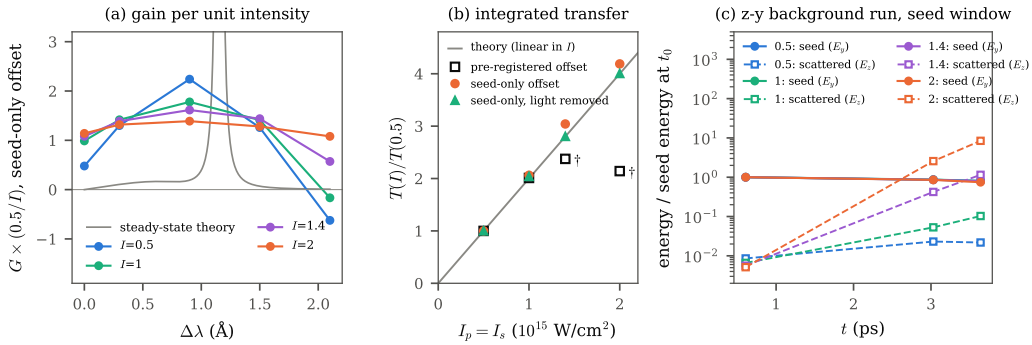


Figure 8: The laser-intensity axis at $\mathbf{B} = 0$ (new reduced-scale runs). (a) Envelope gain per unit intensity (seed-only offset) and the calibrated steady-state theory, which is the same for every intensity in these units. (b) Integrated transfer relative to the anchor intensity for the pre-registered offset, the seed-only offset, and the seed-only offset with the scattered light removed; line: linear-in-I prediction; †: comparison invalid. (c) Energy in the seed's spectral window of the z-y background run, split into the y-polarized seed and z-polarized light (pump light scattered into the seed direction), relative to the seed energy at t_0. Everything except the pre-registered points in (b) is post-hoc analysis.

At 2×10^{15} W/cm^2 the pre-registered analysis finds $T_{\mathrm{PIC}} = 5.36 \pm 1.78$ against $C_f T_{\mathrm{V}} = 10.0$, i.e. $\Delta = +0.46 \pm 0.25$: the cell breaks (illustrative), the deterministic rule attributes the failure to the nonlinear ion response (B-ii; the ion bounce frequency in the driven wave exceeds the linear damping rate 33.9-fold), and the pre-registered test of sublinear transfer (M2) gives $p = 0.0339$, Holm-adjusted 0.102. That last number sits on the boundary of the family-wise level: Eq. 6 divides by the bootstrapped calibration constant, the bootstrap distribution of Δ for this cell is heavy-tailed, and its standard deviation is 0.25 for the pre-registered bootstrap seed but 0.21–0.22 for four other seeds, which put M2 at the family-wise level (Holm-adjusted p = 0.045–0.050; no cell label changes with the seed, `logs/bootstrap_seed_sensitivity.json`).

This result does not survive inspection, and the inspection was not pre-registered. The z-y background run, whose seed energy defines the offset, *gains* energy at this intensity ($g = +1.72$, against -0.17 at 0.5×10^{15} W/cm^2). Splitting the seed's spectral window by field component (Fig. 8c) shows why: the y-polarized seed loses energy as at low intensity (-0.22 versus -0.19), while z-polarized light in the same window—pump light scattered toward the seed direction and amplified from noise, that is, Brillouin side-scatter in the geometry of the crossing—grows to 92% of the window energy by t_2 (3% at 0.5 and 11% at 10^{15} W/cm^2). The pre-registered measure sums all field components, so at 2×10^{15} W/cm^2 the offset measures scattered light rather than the seed's losses. Before the 10^{15} W/cm^2 background run existed we stated three quantitative predictions of this diagnosis for it (`logs/posthoc_prediction_I1.json`, hashed in the plan log): a seed-only offset within 0.1 of its low-intensity value, a scattered share of the window between 5% and 92%, and an upward bias of the all-component offset larger than 0.05. All three hold (-0.18, 11%, $+0.08$), and the same test, stated again before the 1.4×10^{15} W/cm^2 background run (below), passed as well (-0.19, 59%, $+0.64$), so the diagnosis is supported on two fresh cells. With the seed-only offset the transfer at 2×10^{15} W/cm^2 is $T = 10.8$ ($\Delta = -0.07$); removing the scattered energy from the z-z windows as well, which over-corrects because seed and scattered light compete for the same pump, gives 10.1 ($\Delta = -0.01$). The windowed transfer thus scales linearly with intensity to within a few per cent (Fig. 8b).

Because the pre-registered labels change between 10^{15} and 2×10^{15} W/cm^2, the pre-registered refinement rule added a gain curve at 1.4×10^{15} W/cm^2. It holds under the pre-registered analysis ($\Delta = +0.15 \pm 0.28$), but its background window already contains 59% scattered light, which biases Δ upward: with the seed-only offset $\Delta = -0.11$, with the light removed -0.01. The pre-registered boundary between 1.4 and 2×10^{15} W/cm^2 is therefore an artifact of the background measurement. The physical changes along the axis are continuous instead:

from 0.5 to 1, 1.4 and $2 \times 10^{15}\,\mathrm{W/cm^2}$ the scattered share of the background window grows as 3, 11, 59 and 92%, the edge-to-peak ratio of the gain curve as 0.21, 0.56, 0.67 and 0.82, and the pump depletion at the seed exit as 1.3, 2.0, 6.6 and 20.1%.

That agreement does not test the theory, however. The gain curve at $2 \times 10^{15}\,\mathrm{W/cm^2}$ is flat across the window (Fig. 8a): its edges reach 0.82 of the peak (0.21 at 0.5 and 0.56 at $10^{15}\,\mathrm{W/cm^2}$), so the response is not contained in the detuning window and the sum rule of Sec. 2.3, which protects only the full integral of a linear response, does not apply. The gain grows monotonically in time at every detuning instead of following the transient of the linear response (Fig. 7, bottom row), and the pump is depleted by up to 20.1% at the seed exit, so that with the corrected offset prerequisite A3 fails as well. The cell lies outside the domain in which a linear theory can be compared with these simulations; the B-ii attribution and the evidence M2 gives for sublinear transfer are artifacts of the background measurement and we do not count them as support for assumption (ii). Two of the failure modes—scattered light in the seed's spectral window and a response that outgrows the detuning window—were not among the pre-registered comparison prerequisites; both are cheap to check (polarization-resolved window energies, gain at the window edges) and belong in any high-intensity CBET benchmark. The authors of the archived study chose $0.5 \times 10^{15}\,\mathrm{W/cm^2}$ because noise-seeded instabilities began to compete when the pump intensity was doubled [Shi and Moody, 2026]; our runs quantify how such light corrupts the gain measurement and where.

5.8 Mechanism attribution

Table 3 collects every boundary segment we found or tested, with the attributed mechanism, its category (comparison prerequisite or physical assumption of the theory), whether the hypothesis was stated before the residuals were seen, and its statistical support; Fig. 9 shows the underlying diagnostics. The only *established* attribution is the quasi-steady-state prerequisite for pointwise comparisons (M5). The pre-registered test of the nonlinear ion response, M2 (sublinear transfer at $2 \times 10^{15}\,\mathrm{W/cm^2}$), is borderline—Holm-adjusted $p = 0.102$ for the pre-registered bootstrap seed, 0.045–0.050 for other seeds—and in any case confounded by the contaminated background measurement diagnosed in Sec. 5.7; it is not counted. The theory's physical approximations are thus tested only indirectly: the drive approximation (i) changes predictions by two orders of magnitude less than the threshold (Fig. 9b; M3 not significant); the linear ion response (ii) is consistent with the doubling of the integrated transfer between 0.5 and $10^{15}\,\mathrm{W/cm^2}$, the only intensity step at which the direct (exploratory) diagnostics show an undepleted pump and a clean background measurement, although the pre-registered pump-depletion proxy flags it; and the single-quasimode, unperturbed-orbit response (iii) remains a candidate for the near-perpendicular residual, which is suggestive (M1) but below threshold and, as the failed magnetized validation shows, comparable to the numerical uncertainty of the simulations there. The finite-time variant moves the normalized transfer by less than $\tau/2$ in every cell except the two 30-MG hydrogen cells (Fig. 9c), which is the sum rule at work.

5.9 Compactness

On the primary observable the boundary is compact in the strongest sense within the archived region: no archived cell breaks, so the majority rule ("the theory holds") classifies all 64 archived cells correctly and no axis is needed; the continuous residual is carried by a single field-strength coordinate (Sec. 5.4). Including the new cells, the only primary "breaks" label is the $2 \times 10^{15}\,\mathrm{W/cm^2}$ cell, so the best single-variable rule is trivially a threshold on intensity (leave-one-out accuracy 1.00; max-statistic permutation $p = 0.0605$, not significant for a single label); as that label is a diagnosed artifact of the background measurement, the substantive statement is unchanged. The boundary that the archive *does* contain—the pointwise one of Sec. 5.5—is also compact. In an exploratory analysis (not pre-registered; `logs/pointwise_compactness.json`) with the same decision-stump machinery, orientation alone separates the pointwise failures: "fails if $\theta_{kB} > 68°$" has leave-one-out accuracy 0.90 against a base rate of 0.64 (max-statistic permutation $p < 10^{-4}$), and adding field strength does not improve it (Fig. 10). Field strength and polarization are, by themselves, uninformative (0.60 and 0.64).

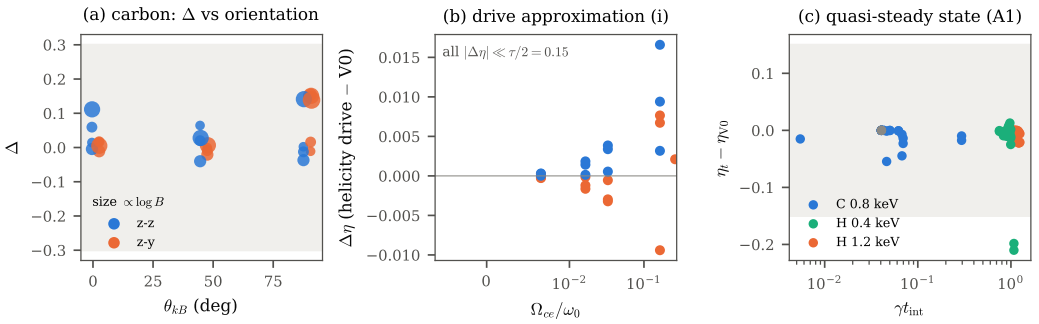


Figure 9: Mechanism diagnostics. (a) Residual versus plasma-wave orientation for carbon (marker size $\propto \log B$). (b) Change of the predicted η when the symmetrized potential drive is replaced by the helicity-retaining drive, versus Ω_{ce}/ω_0. (c) Change of the predicted η when the steady-state response is replaced by the finite-time response at the measurement times, versus $\gamma t_{\rm int}$ (grey band: $\pm\tau/2$).

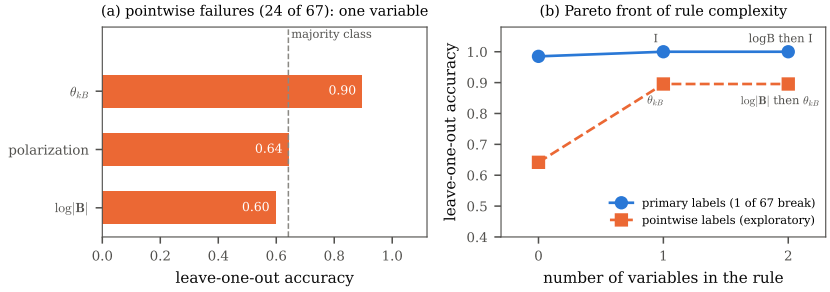


Figure 10: Compactness. (a) Leave-one-out accuracy of single-variable rules for the pointwise ($\Delta\lambda = 0.9$ Å) failure labels of the archived carbon field scans (exploratory; dashed: majority-class rate). (b) Pareto front of rule complexity versus leave-one-out accuracy for the primary labels of all tested cells and for the pointwise labels.

Table 5: Primary results: every cell with a gain curve. Parameters ($|\mathbf{B}|$ [MG], θ_{kB} [deg], n_e [10^{20} cm^{-3}], T_e [keV], I [10^{15} W/cm^2], polarization); family C = archived carbon, $H_{0.4}/H_{1.2}$ = archived hydrogen, C^r = new reduced-scale carbon runs. $C_f T_V$: calibrated first-order Vlasov transfer; $T_{\rm PIC}$: measured transfer (mean ± bootstrap s.d.), both in Å; Δ: agreement metric, Eq. (6). Class: H/B = holds/breaks at $\tau = 0.30$; * established (Holm, FWER 0.05), otherwise illustrative; † thin support (spread from < 3 samples) or no spread. Mechanism: attribution of "breaks" cells; for "holds" cells, any comparison prerequisite that fails (A1 quasi-steady state, A3 pump depletion, A5 signal-to-noise, A6 finite-gain nonlinearity), which makes the label uninformative. §Comparison invalid by the post-hoc diagnosis of Sec. 5.7 (more than half of the background window is scattered light). "valid. cell": the magnetized validation cell of the reduced configuration (failed V2; not classified).

Cell	$(\|\mathbf{B}\|, \theta_{kB}, n_e, T_e, I, \text{pol})$	$C_f T_V$	$T_{\rm PIC}$	Δ	σ_Δ	Class	Mechanism
C	(0, –, 3.5, 0.8, 0.5, z-z)	2.95	2.95±0.48	0 (def.)	–	anchor	–
C	(0, –, 3.5, 0.8, 0.5, z-y)	0.00	0.00±0.68	–	–	offset	–
C	(2, 1, 3.5, 0.8, 0.5, z-z)	2.95	2.90±0.49	+0.01	0.15	H	–
C	(5, 1, 3.5, 0.8, 0.5, z-z)	2.92	2.74±0.49	+0.06	0.15	H	–
C	(10, 1, 3.5, 0.8, 0.5, z-z)	2.82	2.83±0.48	+0.00	0.14	H	–
C	(50, 1, 3.5, 0.8, 0.5, z-z)	1.67	1.35±0.52	+0.11	0.15	H	–
C	(2, 1, 3.5, 0.8, 0.5, z-y)	0.01	-0.04±0.58	+0.02	0.22	H	prereq. A5
C	(5, 1, 3.5, 0.8, 0.5, z-y)	0.04	-0.01±0.50	+0.02	0.20	H	prereq. A5
C	(10, 1, 3.5, 0.8, 0.5, z-y)	0.14	0.17±0.49	-0.01	0.18	H	prereq. A5
C	(50, 1, 3.5, 0.8, 0.5, z-y)	1.45	1.44±0.47	+0.01	0.12	H	–

Cell	$(\lvert\mathbf{B}\rvert, \theta_{kB}, n_e, T_e, I, \mathrm{pol})$	$C_f T_\mathrm{V}$	T_PIC	Δ	σ_Δ	Class	Mechanism
C	(2, 46, 3.5, 0.8, 0.5, z-z)	2.89	2.70±0.48	+0.06	0.14	H	–
C	(5, 46, 3.5, 0.8, 0.5, z-z)	2.62	2.56±0.48	+0.02	0.13	H	–
C	(10, 46, 3.5, 0.8, 0.5, z-z)	1.96	2.08±0.47	-0.04	0.12	H	–
C	(50, 46, 3.5, 0.8, 0.5, z-z)	1.57	1.49±0.48	+0.03	0.13	H	–
C	(2, 46, 3.5, 0.8, 0.5, z-y)	0.06	0.06±0.49	+0.00	0.19	H	prereq. A5
C	(5, 46, 3.5, 0.8, 0.5, z-y)	0.33	0.36±0.48	-0.01	0.17	H	–
C	(10, 46, 3.5, 0.8, 0.5, z-y)	0.99	1.05±0.47	-0.02	0.13	H	–
C	(50, 46, 3.5, 0.8, 0.5, z-y)	1.50	1.48±0.47	+0.01	0.12	H	–
C	(2, 89, 3.5, 0.8, 0.5, z-z)	2.82	2.81±0.49	+0.00	0.15	H	–
C	(5, 89, 3.5, 0.8, 0.5, z-z)	2.29	2.33±0.45	-0.01	0.12	H	–
C	(10, 89, 3.5, 0.8, 0.5, z-z)	1.18	1.29±0.45	-0.04	0.12	H	–
C	(50, 89, 3.5, 0.8, 0.5, z-z)	0.87	0.45±0.48	+0.14	0.16	H	–
C	(2, 89, 3.5, 0.8, 0.5, z-y)	0.07	0.10±0.49	-0.01	0.18	H	prereq. A5
C	(5, 89, 3.5, 0.8, 0.5, z-y)	0.36	0.31±0.48	+0.02	0.17	H	–
C	(10, 89, 3.5, 0.8, 0.5, z-y)	0.72	0.26±0.48	+0.15	0.17	H	prereq. A5
C	(50, 89, 3.5, 0.8, 0.5, z-y)	0.80	0.35±0.41	+0.15	0.14	H	prereq. A5
C	(80, 89, 3.5, 0.8, 0.5, z-y)	0.77	0.36±0.48	+0.14	0.17	H	prereq. A5
$H_{0.4}$	(0, –, 3.0, 0.4, 1, z-z)	4.69	4.69±0.05	0 (def.)	–	anchor	–
$H_{0.4}$	(0, –, 3.0, 0.4, 1, z-y)	0.00	0.02±0.06	–	–	offset	–
$H_{0.4}$	(2.5, 72, 3.0, 0.4, 1, z-z)	4.61	4.01±0.05	+0.13	0.00	$H^\dagger$	prereq. A3
$H_{0.4}$	(5, 72, 3.0, 0.4, 1, z-z)	4.39	4.09±0.05	+0.06	0.00	$H^\dagger$	–
$H_{0.4}$	(7.5, 72, 3.0, 0.4, 1, z-z)	4.09	3.72±0.05	+0.08	0.00	$H^\dagger$	prereq. A6
$H_{0.4}$	(10, 72, 3.0, 0.4, 1, z-z)	3.76	3.06±0.05	+0.15	0.00	$H^\dagger$	prereq. A6
$H_{0.4}$	(12.5, 72, 3.0, 0.4, 1, z-z)	3.40	3.03±0.05	+0.08	0.00	$H^\dagger$	prereq. A6
$H_{0.4}$	(15, 72, 3.0, 0.4, 1, z-z)	3.07	2.49±0.05	+0.12	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(17.5, 72, 3.0, 0.4, 1, z-z)	2.79	2.47±0.05	+0.07	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(20, 72, 3.0, 0.4, 1, z-z)	2.58	2.22±0.06	+0.08	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(22.5, 72, 3.0, 0.4, 1, z-z)	2.45	1.82±0.06	+0.13	0.01	$H^\dagger$	–
$H_{0.4}$	(25, 72, 3.0, 0.4, 1, z-z)	2.39	1.62±0.06	+0.16	0.01	$H^\dagger$	–
$H_{0.4}$	(27.5, 72, 3.0, 0.4, 1, z-z)	2.38	1.77±0.06	+0.13	0.01	$H^\dagger$	–
$H_{0.4}$	(30, 72, 3.0, 0.4, 1, z-z)	2.40	1.69±0.06	+0.15	0.01	$H^\dagger$	prereq. A1
$H_{0.4}$	(2.5, 72, 3.0, 0.4, 1, z-y)	0.06	0.21±0.06	-0.03	0.01	$H^\dagger$	–
$H_{0.4}$	(5, 72, 3.0, 0.4, 1, z-y)	0.24	0.41±0.06	-0.04	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(7.5, 72, 3.0, 0.4, 1, z-y)	0.51	0.54±0.06	-0.01	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(10, 72, 3.0, 0.4, 1, z-y)	0.84	1.11±0.06	-0.06	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(12.5, 72, 3.0, 0.4, 1, z-y)	1.19	1.28±0.06	-0.02	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(15, 72, 3.0, 0.4, 1, z-y)	1.52	1.41±0.06	+0.02	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(17.5, 72, 3.0, 0.4, 1, z-y)	1.80	1.53±0.06	+0.06	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(20, 72, 3.0, 0.4, 1, z-y)	2.01	1.59±0.06	+0.09	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(22.5, 72, 3.0, 0.4, 1, z-y)	2.14	1.56±0.06	+0.12	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(25, 72, 3.0, 0.4, 1, z-y)	2.21	1.87±0.06	+0.07	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(27.5, 72, 3.0, 0.4, 1, z-y)	2.22	1.66±0.06	+0.12	0.01	$H^\dagger$	prereq. A6
$H_{0.4}$	(30, 72, 3.0, 0.4, 1, z-y)	2.20	1.82±0.06	+0.08	0.01	$H^\dagger$	prereq. A1
$H_{1.2}$	(0, –, 3.0, 1.2, 1, z-z)	2.46	2.46±0.14	0 (def.)	–	anchor	–
$H_{1.2}$	(0, –, 3.0, 1.2, 1, z-y)	0.00	0.01±0.08	–	–	offset	–
$H_{1.2}$	(2.5, 72, 3.0, 1.2, 1, z-z)	2.42	2.48±0.11	-0.02	0.07	$H^\dagger$	–
$H_{1.2}$	(5, 72, 3.0, 1.2, 1, z-z)	2.30	2.35±0.11	-0.02	0.07	$H^\dagger$	–
$H_{1.2}$	(7.5, 72, 3.0, 1.2, 1, z-z)	2.11	2.21±0.11	-0.04	0.07	$H^\dagger$	–
$H_{1.2}$	(10, 72, 3.0, 1.2, 1, z-z)	1.91	2.15±0.11	-0.10	0.06	$H^\dagger$	–
$H_{1.2}$	(12.5, 72, 3.0, 1.2, 1, z-z)	1.72	1.80±0.12	-0.03	0.06	$H^\dagger$	–
$H_{1.2}$	(15, 72, 3.0, 1.2, 1, z-z)	1.54	1.86±0.11	-0.13	0.06	$H^\dagger$	–
$H_{1.2}$	(17.5, 72, 3.0, 1.2, 1, z-z)	1.39	1.31±0.17	+0.04	0.07	$H^\dagger$	–
$H_{1.2}$	(20, 72, 3.0, 1.2, 1, z-z)	1.28	1.54±0.10	-0.10	0.05	$H^\dagger$	–
$H_{1.2}$	(2.5, 72, 3.0, 1.2, 1, z-y)	0.03	0.12±0.07	-0.04	0.03	$H^\dagger$	–
$H_{1.2}$	(5, 72, 3.0, 1.2, 1, z-y)	0.13	0.38±0.08	-0.10	0.03	$H^\dagger$	–
$H_{1.2}$	(7.5, 72, 3.0, 1.2, 1, z-y)	0.27	0.48±0.08	-0.09	0.03	$H^\dagger$	–
$H_{1.2}$	(10, 72, 3.0, 1.2, 1, z-y)	0.44	0.70±0.08	-0.10	0.03	$H^\dagger$	–
$H_{1.2}$	(12.5, 72, 3.0, 1.2, 1, z-y)	0.62	0.68±0.07	-0.02	0.03	$H^\dagger$	prereq. A6
$H_{1.2}$	(17.5, 72, 3.0, 1.2, 1, z-y)	0.94	1.14±0.11	-0.08	0.05	$H^\dagger$	prereq. A6

Cell	$(\lvert\mathbf{B}\rvert, \theta_{kB}, n_e, T_e, I, \text{pol})$	$C_f T_V$	T_{PIC}	Δ	σ_Δ	Class	Mechanism
$H_{1.2}$	(20, 72, 3.0, 1.2, 1, z-y)	1.04	1.25±0.12	-0.09	0.05	H[†]	–
C^r	(0, –, 3.5, 0.8, 0.5, z-z)	2.50	2.50±0.43	0 (def.)	–	anchor	–
C^r	(0, –, 3.5, 0.8, 0.5, z-y)	0.00	– (1 pt)	–	–	offset	–
C^r	(10, 89, 3.5, 0.8, 0.5, z-z)	1.00	1.93±0.43	-0.37	0.14	valid. cell	–
C^r	(0, –, 3.5, 0.8, 1, z-y)	0.00	– (1 pt)	–	–	offset	–
C^r	(0, –, 3.5, 0.8, 1, z-z)	5.01	5.04±0.60	-0.01	0.31	H	prereq. A3
C^r	(0, –, 3.5, 0.8, 1.4, z-y)	0.00	– (1 pt)	–	–	offset	–
C^r	(0, –, 3.5, 0.8, 1.4, z-z)	7.01	5.95±0.86	+0.15	0.28	H[§]	invalid[§]
C^r	(0, –, 3.5, 0.8, 2, z-y)	0.00	– (1 pt)	–	–	offset	–
C^r	(0, –, 3.5, 0.8, 2, z-z)	10.02	5.36±1.78	+0.46	0.25	B[§]	B-ii (rule)[§]

Compactness summary: 67 tested cells, 1 "breaks"; majority-class accuracy 0.99; best one-variable rule: I (leave-one-out accuracy 1.00, permutation $p = 0.0605$); best two-variable rule: logB→I (leave-one-out 1.00).

6 Discussion

6.1 Failure analysis

Unsupported or fuzzy cells. Every archived label is a "holds" label, and every one of them is illustrative. In carbon the reason is the shared background offset, whose pre-registered uncertainty inherits an analysis artifact of the z-y, $\mathbf{B} = 0$ cell (Appendix B); in hydrogen the archived spreads come from two measurement times (1.2 keV) or none (0.4 keV). The weakness is therefore not a large residual but a thin error model. A further, unquantified uncertainty concerns the near-perpendicular magnetized cells: the reduced-resolution validation (Sec. 5.6) shows that their simulated transfer is far more sensitive to resolution than the unmagnetized transfer ($\Delta\eta = +0.33$ between 12 and 20 cells per λ_0 at 10 MG), and the archived spreads, which sample measurement times and integration fractions only, do not contain this numerical uncertainty. Whether the paper resolution is converged there is unknown, so the near-perpendicular residuals of Sec. 5.4 carry this caveat. The sensitivity analysis shows that the same residuals would establish 11 of the 25 carbon labels with a less conservative offset error; we did not adopt it because the choice was made after seeing residuals. Of the new cells, the 10^{15} W/cm^2 label is flagged by the pre-registered pump-depletion proxy although the pump is measured to be undepleted, and the 2×10^{15} W/cm^2 label is invalid by the post-hoc diagnosis of Sec. 5.7; the new runs therefore add no established label either.

Attributions that did not work. The one cell that breaks, at 2×10^{15} W/cm^2, was attributed by the deterministic rule to the nonlinear ion response (B-ii), and the matching test M2 is borderline (Holm-adjusted $p = 0.102$, near 0.05 for other bootstrap seeds); both are artifacts of a background measurement contaminated by scattered light, and the pre-registered indicators would not have caught it (A3 did not fire because the contaminated offset lowered the measured gain). The failure of the reduced configuration at 10 MG could not be attributed beyond excluding noise, bulk heating and background. Because no archived cell breaks, the rule had nothing to attribute in the archive; the near-perpendicular residual pattern (theory over-predicting both polarizations by ≈ 0.15 at ≥ 10 MG) is the closest the archive comes to a boundary, and the pre-registered test that singles it out (M1) is not significant. The drive approximation (i) is irrelevant in the explored range: replacing the symmetrized potential drive by the helicity-retaining form changes the predicted η by at most 0.017 even at $\Omega_{ce}/\omega_0 = 0.26$, an order of magnitude below the residuals, and it reduces $|\Delta|$ in only 1 of the 7 cells with $\Omega_{ce}/\omega_0 \geq 0.1$ (M3, $p = 0.992$). The pre-registered seed-angle hypothesis (A7) is also ruled out as the source of the near-perpendicular residual: averaging the theory over the seed's angular spectrum changes η there by at most 0.027 (0.058 for a 7° spectrum), and a direct check of the new magnetized runs shows the simulated seed polarization rotating along x as the cold-plasma eigenmodes predict, to ≤ 0.05 rad rms over a Faraday phase of 4.0 rad (`logs/faraday_check.json`), so the laser-eigenmode part of the theory is not the culprit either. The finite-time variant changes carbon η by at most 0.055,

Table 3: Boundary segments and their attribution.

Segment	Status	Attributed mechanism	Pre-stated?	Support
Pointwise (0.9 Å) comparison, carbon field scans	$\lvert\Delta_{0.9}\rvert > \tau$ in 24/73 points, mainly $\theta_{kB} \gtrsim 68°$	A1 quasi-steady state (comparison validity)	yes (A1, M5)	M5: 20/25 cells, Holm $p = 0.00815$
Integrated transfer, all 64 archived cells	no crossing ($\lvert\Delta\rvert \leq 0.16$)	none required	yes	descriptive (labels illustrative)
Carbon $\theta_{kB} = 89°$, ≥ 10 MG (largest residuals)	no crossing ($\Delta \leq +0.15$)	unattributed; candidates B-iii, A7 (A7 shift of $\eta \leq 0.03$), finite field-line length, PIC convergence	B-iii, A7 yes; others post hoc	M1 Holm $p = 0.25$ (n.s.)
Reduced configuration at 10 MG, 89° (validation)	V2 failed: $\Delta\eta = +0.33 \pm 0.18$	numerical (resolution); not noise, heating or background	validation yes; diagnosis post hoc	seeds, pilots, pump-off run; 14 cells/λ_0 closes 23%, 23% of the gap (2 detunings)
Absolute level at $\mathbf{B} = 0$	calibrated, $C_f = 1.40$–2.04	unattributed (window/transient < 10%; coupling at $\Delta\lambda = 0$ not described by the linear response, post hoc)	partly	descriptive
New run: $B = 0$ MG, $I = 1$	holds-ill., $\Delta = -0.01 \pm 0.31$	rule: none (holds); post hoc: scattered light is 11% of the offset window, seed-only offset gives $\Delta = -0.06$ ($\Delta = -0.03$ with the light removed)	rule yes; diagnosis post hoc	Holm $p = 1$
New run: $B = 0$ MG, $I = 1.4$	holds-ill., $\Delta = +0.15 \pm 0.28$	rule: none (holds); post hoc: scattered light is 59% of the offset window, seed-only offset gives $\Delta = -0.11$ ($\Delta = -0.01$ with the light removed)	rule yes; diagnosis post hoc	Holm $p = 1$
New run: $B = 0$ MG, $I = 2$	breaks-ill., $\Delta = +0.46 \pm 0.25$	rule: B-ii nonlinear ion response; post hoc: scattered light is 92% of the offset window, seed-only offset gives $\Delta = -0.07$ ($\Delta = -0.01$ with the light removed)	rule yes; diagnosis post hoc	Holm $p = 1$; M2 Holm $p = 0.102$ (confounded)

Table 4: Pre-registered mechanism tests (family-wise error controlled by Holm's procedure over the evaluable tests).

Test	Hypothesis	n	p	p_{Holm}	Decision
M1 (B-iii)	$\lvert\Delta\rvert$ larger at $\theta_{kB} \approx 89°$ than at 0°/46° (matched $\lvert\mathbf{B}\rvert$, pol.)	8	0.125	0.25	not rejected
M2 (B-ii)	transfer at 2×10^{15} W/cm^2 below the calibrated linear-in-I prediction	1	0.0339	0.102	not rejected‡
M3 (B-i)	helicity-retaining drive reduces $\lvert\Delta\rvert$ where $\Omega_{ce}/\omega_0 \geq 0.1$	7	0.992	0.992	not rejected
M5 (A1)	single-detuning residual exceeds integrated residual where $\gamma t_{\text{int}} < 1$	25	0.00204	0.00815	reject H_0

‡Confounded: a post-hoc diagnosis shows that the background run at this intensity measures scattered pump light; with the seed-only offset the transfer is not sublinear (Sec. 5.7). Not counted as support for B-ii.

confirming that the integrated observable is insensitive to the interaction time. The linear ion ponderomotive term (assumption ii in its linear form) changes R by $\lesssim Zm_e/m_i \sim 3 \times 10^{-4}$.

Irrelevant axes. Polarization and electron temperature carry no residual signal ($\rho = -0.21$ and -0.16); within each family the residual is organized by field strength alone. Threshold changes ($\tau = 0.2$, 0.4) do not change a single archived label, and neither does the bootstrap seed: with 5 seeds the analysis always finds 0 established labels and the same single "breaks" cell. A post-hoc criterion for window containment (gain at the window edge above half the peak) was considered and dropped because it would have been chosen with the 10^{15} W/cm^2

curve in view; the edge-to-peak ratios are reported instead (0.21, 0.56, 0.67 and 0.82 at 0.5, 1, 1.4 and $2 \times 10^{15}\,\mathrm{W/cm^2}$).

The coherent pattern. Every apparent failure we found is a failure of the comparison rather than of the theory: pointwise comparisons fail the quasi-steady-state prerequisite wherever the ion-acoustic response is weakly damped (all carbon cells, established by M5); the absolute coupling is matched only after calibration; the reduced-resolution simulations fail to reproduce the paper-resolution transfer near perpendicularity; and at high intensity scattered light corrupts the background measurement while the response outgrows the detuning window. Once the comparison is made on the sum-rule-protected observable, inside the regime where it is valid, the theory's physical approximations are not visibly violated anywhere; the residual that remains grows smoothly with magnetization and is largest exactly where the simulations themselves are most sensitive to resolution.

6.2 Computational cost and the value of more simulations

The new simulations comprise 34 EPOCH runs at 12 cells per λ_0 (85 min each on 15 MPI ranks, including one pump-off diagnostic) and 2 at 14 cells per λ_0 (125 min, in two restart segments), 52.2 h of wall-clock in total, one run at a time; at the paper resolution each run would have cost ≈ 8.9 h on 16 ranks. The steady-state theory needs ≈ 26 s per cell on one core for an 8001-point detuning integral (all baseline cells: 0.6 core-h), and the diagnostic variants (two drive forms, finite-time response, seed-angle average) a further 1.9 core-h. One additional gain-curve cell costs five reduced-scale runs, i.e. ≈ 7.1 h of the 16-core budget, against under a minute for the theory: the theory is $\sim 1 \times 10^4\times$ cheaper per cell even before counting the uncertainty runs. Would five more runs have extended the map substantially? Not in the magnetized directions: the reduced configuration failed its magnetized validation, so a new magnetized cell needs either the paper resolution (a five-detuning gain curve costs ≈ 45 h) or a validated finer configuration (one rung of the resolution ladder alone costs ≈ 23 h). Along the intensity axis five runs buy one more gain curve, and above $10^{15}\,\mathrm{W/cm^2}$ such a curve locates the breakdown of the comparison rather than testing the theory. The largest return per run would come instead from (a) two near-perpendicular magnetized runs at or above the paper resolution, which would put a number on the convergence uncertainty that now limits the interpretation of the largest archived residuals, and (b) repeats of archived carbon cells with further seeds, which would shrink the dominant offset uncertainty and are the only route to established labels in the archived map.

6.3 Relation to previous work

CBET theory, models and saturation. Linear CBET theory goes back to stimulated Brillouin scattering by a driven ion-acoustic quasimode [Randall et al., 1981, Kruer et al., 1996]; it underlies the ray-based and inline models used in ICF design [Michel et al., 2009, 2010, Follett et al., 2018, Colaïtis et al., 2016] and is routinely calibrated against experiments by multipliers or saturation clamps [Strozzi et al., 2017, Kirkwood et al., 2013]. Kinetic simulations have mapped the departures from linear transfer at high intensity: ion heating and pump depletion [Michel et al., 2012, Hansen et al., 2022], trapping-induced detuning [Nguyen et al., 2021], and time-dependent saturation [Seaton et al., 2022, Yin et al., 2023]. Ps-to-ns simulations show that a quasi-steady state is reached only after tens of picoseconds [Xu et al., 2026], and plane-wave gains overestimate transfer between smoothed, focused beams because angular spread detunes the resonance [Oudin et al., 2025]. Polarization-resolved (unmagnetized) CBET has recently been formulated analytically [Michel et al., 2026]. Those studies compare theory and simulation at fixed conditions and at a given detuning; our use of a sum-rule-protected integrated observable is designed precisely for the opposite limit of weakly damped waves and short interaction times, where a pointwise comparison is not meaningful.

Magnetized laser–plasma interactions. Magnetization modifies parametric processes through the participating waves and their coupling [Stenflo, 1970, Liu and Tripathi, 1986, Cohen, 1987, Shi et al., 2017, Shi, 2019]. Kinetic theory, PIC simulations and experiments have addressed magnetized Raman scattering [Boyd and Rankin, 1985, Winjum et al., 2018,

Bolaños et al., 2025, Lee et al., 2026], multi-beam instabilities in magnetized underdense plasmas [Lacoste et al., 2026], magnetized two-plasmon decay [Li et al., 2025], laser amplification and absorption in strong fields [Edwards et al., 2019, Manzo et al., 2022], benchmarks of magnetized three-wave coupling against PIC [Shi, 2023], and experiments in magnetized underdense plasmas [Yao et al., 2023]; reviews summarize the regimes [Shi et al., 2018]. The MagCBET theory we test [Moody and Shi, 2026] combines a low-frequency magnetized ponderomotive force [Lee and Parks, 1983, Karpman and Shagalov, 1982] with the fully magnetized kinetic response [Stix, 1992, Bernstein, 1958], and the PIC study of Ref. [Shi and Moody, 2026] established its qualitative trends. First experiments on OMEGA have so far been reported only in conference abstracts [Moody et al., 2024, Pollock et al., 2024, Moody et al., 2025]. To our knowledge no previous work has delimited where its quantitative predictions hold, and we found no independent test of the theory published since 2025.

Validation methodology. Our procedure follows the verification-and-validation practice of fixing the validation metric, the acceptance threshold and the domain before comparison [Oberkampf and Roy, 2010], uses equivalence-type tests with family-wise error control [Schuirmann, 1987, Holm, 1979], and a max-statistic permutation test for model selection [Westfall and Young, 1993]. The sum rule we rely on is the causality (Kramers–Kronig) property of linear response [Kubo, 1957]; using it to make a steady-state plasma theory comparable with a finite-duration simulation is, to our knowledge, new in the laser–plasma context.

6.4 Implications for ICF practice

What the linear theory can be trusted for. Within the explored conditions (dense, $T_e = 4T_i$ carbon and hydrogen plasmas, $n_e/n_c \approx 0.03$–0.04, crossing angles 24°–30°) the magnetic modification factor $\eta(|\mathbf{B}|, \theta_{kB}, \text{polarization})$ of the first-order Vlasov theory can be applied as a multiplicative correction to a CBET rate that has been calibrated without field, with errors $\lesssim 0.16$ of the unmagnetized transfer up to 80 MG in carbon and 30 MG in hydrogen, and the integrated transfer scales linearly with the pump intensity at least up to $10^{15}\,\mathrm{W/cm^2}$ in carbon. This covers, with a wide margin, the self-generated fields near hohlraum entrance holes ($\sim 10^2\,\mathrm{T} = 1\,\mathrm{MG}$) [Farmer et al., 2017, Pearcy et al., 2025], the imposed fields of present experiments and ignition-class magnetized hohlraum designs (5–75 T) [Moody et al., 2022, Strozzi et al., 2024, Djordjević et al., 2026, Walsh et al., 2025], and the $> 500\,\mathrm{T}$ fields already produced inside coil-driven hohlraums [Zhang et al., 2025]. At the fields of current NIF experiments the predicted η differs from one mainly through Faraday rotation of the beams' polarization over millimetre paths [Moody and Shi, 2026], which the theory captures.

When to fall back to kinetic simulation. (a) The absolute coupling: the theory under-predicts the unmagnetized transfer of the 2D simulations by a factor 1.40–2.04, so the calibration that CBET models already perform [Strozzi et al., 2017, 2024] remains necessary. (b) Near-perpendicular geometry at $\gtrsim 10\,\mathrm{MG}$ with polarizations that Faraday rotation can mix: here the theory over-predicts cross-polarized coupling by up to 0.15 of the unmagnetized transfer—below our threshold, but the largest error in the map, and where the simulations themselves are most sensitive to resolution. (c) Intensities at which noise-seeded scattering and pump depletion set in (here above $\sim 10^{15}\,\mathrm{W/cm^2}$ in 2D carbon slabs): the response broadens beyond the linear line shape, and the simulations no longer test the linear theory with standard diagnostics. (d) Line shapes and time-resolved transfer in weakly damped plasmas: the steady-state gain at a given detuning is unreliable whenever the ion-acoustic damping time exceeds the interaction time ($\gamma t \lesssim 1$); only detuning-integrated (or detuning-averaged, e.g. broadband-smoothed) transfer is protected. Longer simulations reach a quasi-steady state only after tens of picoseconds, by which time trapping and harmonics may already act [Xu et al., 2026].

How to validate MagCBET codes. Benchmarks of CBET models against short PIC runs should use detuning-integrated transfer, or interaction times longer than the ion-acoustic damping time; otherwise apparent failures of the theory are failures of the comparison. For magnetized cases the benchmark should also report the plasma-wave orientation θ_{kB} with the

refraction of the beams taken into account: in the archived geometry the nominal 88° case is at 89.1°, and near perpendicularity this changes the predicted electron response substantially; the finite angular spectrum of focused beams matters for the same reason [Oudin et al., 2025]. Near perpendicularity the simulated magnetized transfer is also far more sensitive to resolution than the unmagnetized transfer—12 instead of 20 cells per wavelength changed η by +0.33—so benchmarks there should demonstrate convergence. At high intensity, finally, gains must be measured on the seed's own polarization: an all-component spectral window collects amplified pump scatter, which in our runs reached 92% of the background window at $2 \times 10^{15}\,\mathrm{W/cm^2}$.

6.5 Limitations

Scope. All simulations are two-dimensional slabs with a single density per species, $T_e = 4T_i$, one crossing angle per family and a focused seed of finite aperture; the map says nothing about lower densities, other temperature ratios, flowing plasmas, or three-dimensional speckled beams [Oudin et al., 2025].

Error model. The archived uncertainties are spreads over two measurement times and three transverse integration fractions, not ensemble statistics; the pre-registered offset error is conservative (Appendix B) and the 0.4-keV hydrogen archive has no spreads. Consequently our "holds" labels are descriptive.

Calibration. Each family is calibrated once at $\mathbf{B} = 0$, so any field-independent error of the theory is absorbed; the absolute under-prediction (factor 1.40–2.04) and the positive simulated gain at $\Delta\lambda = 0$ remain unexplained, as does the systematic offset of the simulated line centre below the linear ion-acoustic resonance (0.9 versus 1.153 Å in carbon).

Reconstruction. The baseline uses the electrostatic limit of the plasma response, transverse orthonormal laser eigenmodes, the pump polarization on the seed axis, and the first-order transfer; our ponderomotive normalization differs by a factor of two from our reading of the published prefactor (immaterial after calibration). Finite-size blurring of the gain curve (Eq. 10 of Ref. [Shi and Moody, 2026]) is not applied.

New simulations. The reduced-scale configuration heats electrons 12 times faster than the paper resolution [Adams et al., 2025]. It reproduces the unmagnetized transfer (V1) but not the near-perpendicular magnetized one (V2, V3), so the new simulations extend the map only at $\mathbf{B} = 0$, and that extension rests on a configuration validated at a single, unmagnetized reference point; the pre-registered resolution ladder that might have rescued the magnetized cells was not executed in full. Along the intensity axis the comparison is informative only up to $10^{15}\,\mathrm{W/cm^2}$, and even there only with an exploratory measurement of pump depletion that overrides the pre-registered proxy; the diagnosis of the high-intensity cell (scattered light in the background window) is post hoc, supported on two fresh cells.

Single seeds per cell. Apart from the two validation cells (three seeds each), every new gain point comes from one random seed; the pooled between-seed scatter is added to its uncertainty but not measured cell by cell.

Pre-registration. Two amendments were made before any residual was computed (first-order transfer; seed angular spread) and a third, a quality-control fallback of the gain extraction, before any residual of the new simulations; the execution order of the new runs was changed once, before any run at a new intensity existed, to complete the intensity axis first; after the magnetized validation failed, the unmagnetized intensity cells were evaluated without the fallback ladder and the magnetized intensity cells were dropped. A disclosed sensitivity analysis and several exploratory diagnostics (listed in Sec. 4), including the scattered-light correction, were added after residuals were seen.

7 Conclusions

We asked where the linear Vlasov theory of magnetized cross-beam energy transfer can replace kinetic simulation. The answer depends first on *what* is compared. For the weakly damped ion-acoustic waves of the archived carbon simulations, gains at a fixed detuning measure a picosecond transient and disagree with any steady-state theory; the detuning-integrated first-order transfer is protected by a causality sum rule and is the observable a steady-state theory can be held to. On that observable, and after a single calibration at zero field, the theory predicts the magnetic modification of the transfer in every archived cell—carbon and hydrogen, 0–80 MG, plasma waves from parallel to nearly perpendicular to the field, both polarizations—to within 0.16 of the unmagnetized transfer. The residual that remains is one-dimensional: it grows with magnetization and is largest where Faraday rotation mixes the polarizations of nearly perpendicular geometries. All apparent failures we found in the archive are failures of comparison prerequisites (finite interaction time, absolute calibration), not of the theory's physical approximations, whose own diagnostics (drive, linear ion response) change the predictions far less than the residuals. New reduced-scale simulations add the laser intensity: the integrated transfer doubles between 0.5 and 10^{15} W/cm^2 as the linear theory predicts, while at 2×10^{15} W/cm^2 the comparison itself fails—noise-seeded scattering corrupts the background measurement, the pump depletes and the response outgrows the detuning window—so that a pre-registered failure attributed to the nonlinear ion response is an artifact. The same campaign showed that near-perpendicular magnetized transfer is far more sensitive to numerical resolution than unmagnetized transfer, which limits how much of the largest archived residual can be charged to the theory. For ICF practice this means that MagCBET magnetic-modification factors can be layered onto calibrated CBET models across the field range of present and planned magnetized experiments, that cross-polarized coupling at near-perpendicular, multi-megagauss conditions deserves kinetic checks, and that MagCBET benchmarks must use integrated transfer or interaction times longer than the ion-acoustic damping time, polarization-resolved gain measurements at high intensity, and demonstrated convergence near perpendicularity.

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A Derivation details of the reconstructed theory

Coupled-mode equations. With real fields $\mathbf{E} = \mathrm{Re}[\mathbf{E}_p e^{-i\omega_0 t} + \mathbf{E}_s e^{-i\omega_1 t}]$, the cold electron quiver velocities are $\mathbf{v}_{p,s} = -i(e/m_e\omega)\mathsf{M}(\omega)\cdot\mathbf{E}_{p,s}$ with

$$\mathsf{M}(\omega) = \begin{pmatrix} \frac{\omega^2}{\omega^2-\Omega^2} & \frac{i\omega\Omega}{\omega^2-\Omega^2} & 0 \\ -\frac{i\omega\Omega}{\omega^2-\Omega^2} & \frac{\omega^2}{\omega^2-\Omega^2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (\text{field frame, } \Omega = -eB_0/m_e). \tag{7}$$

The slow part of $\frac{1}{2}m_e\mathbf{v}^2$ is $\mathrm{Re}[\tilde{U}e^{-i\Delta\omega t}]$ with $\tilde{U} = \frac{1}{2}m_e\mathbf{v}_p\cdot\mathbf{v}_s^*$ in the unmagnetized limit; Eq. (1) is its magnetized generalization through the cold susceptibility at $\bar{\omega}$ [Moody and Shi, 2026]. Treating $\tilde{U}$ as a potential energy acting on electrons only, Poisson's equation gives $\delta\tilde{n}_e = -(\epsilon_0 k^2/e^2)\chi_e(1+\chi_i)\epsilon^{-1}\tilde{U}$. The nonlinear current at the seed frequency is $\mathbf{J}_{\mathrm{NL}} = -\frac{1}{2}e\,\delta\tilde{n}_e^*\mathbf{v}_p$. Projecting the wave equation $\nabla\times\nabla\times\mathbf{E} - (\omega_1^2/c^2)\mathsf{K}\cdot\mathbf{E} = i\omega_1\mu_0\mathbf{J}_{\mathrm{NL}}$ on the seed eigenmode m with a slowly varying envelope gives $dA_m/dx = -(\omega_1\mu_0/2k_m)\,\hat{\mathbf{e}}_m^*\cdot\mathbf{J}_{\mathrm{NL}}e^{-ik_m x}$, which leads to Eq. (3). The same growth rate follows from equating the seed Poynting-flux increase with $-\frac{1}{2}\mathrm{Re}(\mathbf{J}_{\mathrm{NL}}^*\cdot\mathbf{E}_s)$, and the ratio of seed gain to plasma absorption $\frac{1}{2}\Delta\omega\,\mathrm{Im}(\delta\tilde{n}_e^*\tilde{U})$ equals $\omega_1/\Delta\omega$ as required by the Manley–Rowe relations.

First-order transfer. Writing Eq. (3) in the field frame $\mathbf{a} = (A_m e^{ik_m x})$ as $d\mathbf{a}/dx = i(\mathsf{D}+\mathsf{C})\mathbf{a}$ with $\mathsf{D} = \mathrm{diag}(k_m)$ and $\mathsf{C}_{mm'} = \Gamma_m u_m w_{m'}$, first-order perturbation theory gives $\mathbf{a}(x) = e^{i\mathsf{D}x}[\mathbf{a}_0 + i\int_0^x e^{-i\mathsf{D}x'}\mathsf{C}e^{i\mathsf{D}x'}\mathbf{a}_0 dx']$, and integrating $|a_m|^2$ over the slab yields Eq. (5) with $\int_0^L (L-x)e^{i\delta x}dx = (1+i\delta L - e^{i\delta L})/\delta^2$. For a single mode $2g^{(1)} = 2\kappa L = G$ exactly; numerically $g/g^{(1)} \to 1$ as the pump intensity tends to zero for all polarization cases (unit tests).

Finite-time response. For a causal response $R(\omega) = \int_0^\infty g(\tau)e^{i\omega\tau}d\tau$ with real g, the response accumulated a time t after the drive is switched on is $R_t(\omega) = \int_0^t g(\tau)e^{i\omega\tau}d\tau$. Its real and imaginary parts are the Fourier transforms of the even and odd extensions of g restricted to $|\tau| < t$, hence $R_t = R * [\sin(ut)/(\pi u)]$ and $\int \mathrm{Im}R_t\,d\omega = \int \mathrm{Im}R\,d\omega$ for all $t > 0$. Near a weakly damped root Ω of ϵ, $R \simeq \mathrm{Res}/(\omega-\Omega)$ and $R_t = R - \mathrm{Res}\,e^{i(\omega-\Omega)t}/(\omega-\Omega)$, the form used for diagnostic (a); for $\gamma \to 0$ the energy transfer is $\propto \sin(\delta t)/\delta$ with $\delta = \omega - \omega_r$, whose integral over δ equals that of the steady-state Lorentzian. For efficiency the root of each mode pair is computed once per cell (at the window edge); this leaves a local artifact of up to 0.48 in G within a few hundredths of an Å of the resonance but changes the integrated transfer by less than 10^{-4} relative (`logs/finite_time_root_check.txt`). The roots are found by multi-start secant iteration on the Landau-continued ϵ and accepted only

if $0 < \gamma/\omega_r < 0.6$ and $|\epsilon| < 10^{-6}(1 + |\chi_e|)$; when none exists the quasimode is overdamped and $R_t = R$.

Unit tests. The cold eigenmodes reproduce $n^2 = P$ and $n^2 = RL/S$ for perpendicular and $n^2 = R, L$ for parallel propagation to 10^{-10}; the magnetized susceptibilities reproduce the unmagnetized limit ($B = 5\,\mathrm{T}$) to $< 2 \times 10^{-3}$ and the static Boltzmann limit to 10^{-6}; the coupled-mode gain at $B = 0$ reproduces the analytic CBET gain to 10^{-6}; z-y transfer vanishes at $B = 0$; negative detuning gives loss (`logs/vlasov_unit_tests.txt`).

B Background offset and its uncertainty

The background offset g_0 is the detuning-averaged Fourier-integral gain of the z-y, $\mathbf{B} = 0$ cell (-0.234 for carbon, flat across detuning with a scatter of 0.006). Its pre-registered standard error combines that scatter with the documented per-point spreads, using the larger of the Fourier-integral and eigenmode-analysis spreads. For this particular cell the eigenmode-analysis spreads (0.42–0.77) are an artifact: with a y-polarized seed and no field, E_z contains no seed signal, the spectral peak search on E_z returns a spurious wave number, and the one-dimensional filter is centred off the seed line; the archived real-mode gains of this cell are accordingly bimodal (-0.60 and -0.19). The resulting offset uncertainty ($\sigma_{g_0} = 0.109$) dominates the carbon error budget. A sensitivity analysis that uses only the Fourier-integral spreads for the offset (not part of the pre-registered analysis; files `*_sens_fi_only`) reduces the median carbon σ_Δ from 0.15 to 0.09 and yields 11 established "holds" labels among the 25 carbon cells; we report it only to show that the absence of established labels in the primary analysis is driven by this conservative choice, not by large residuals.